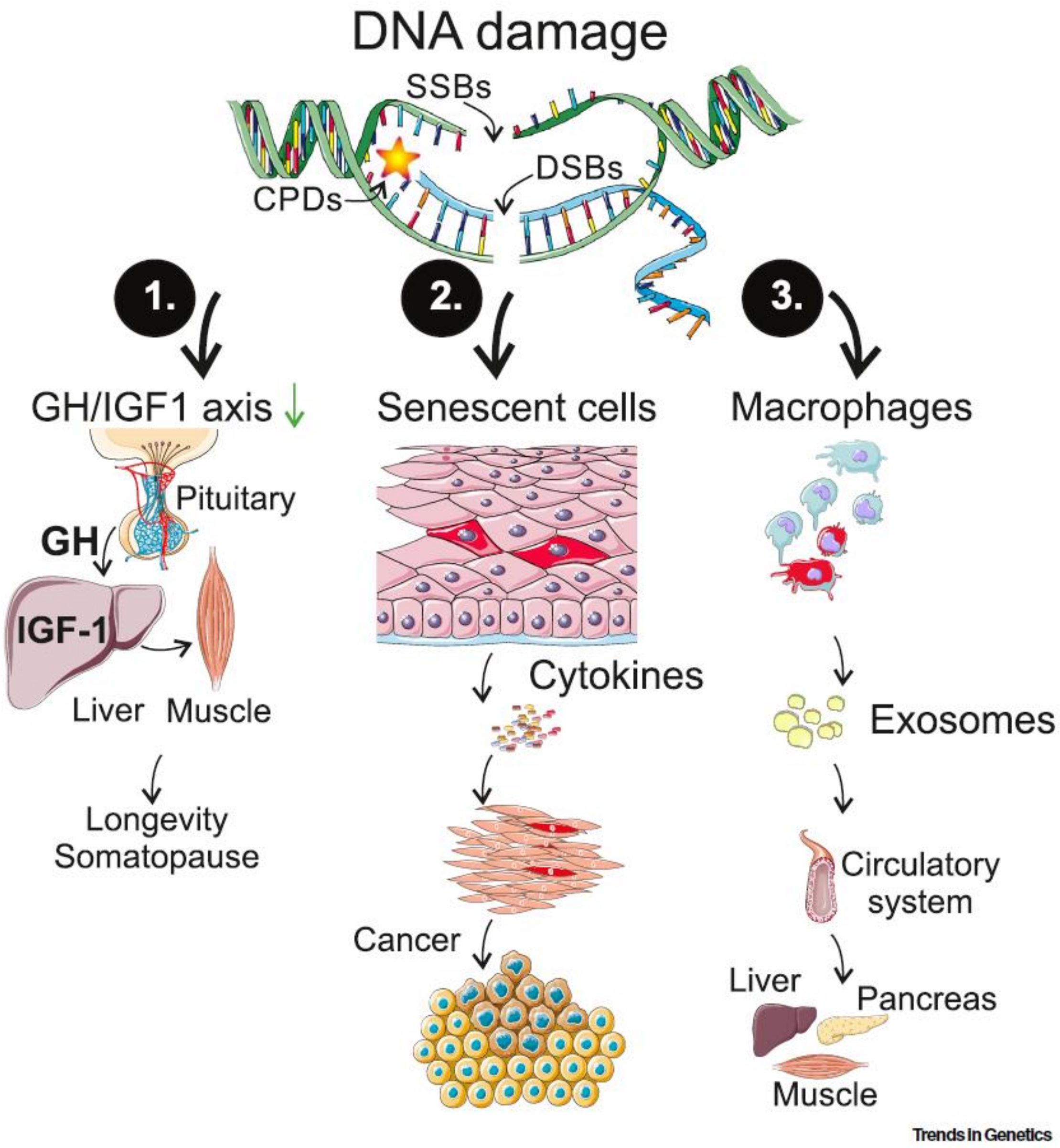

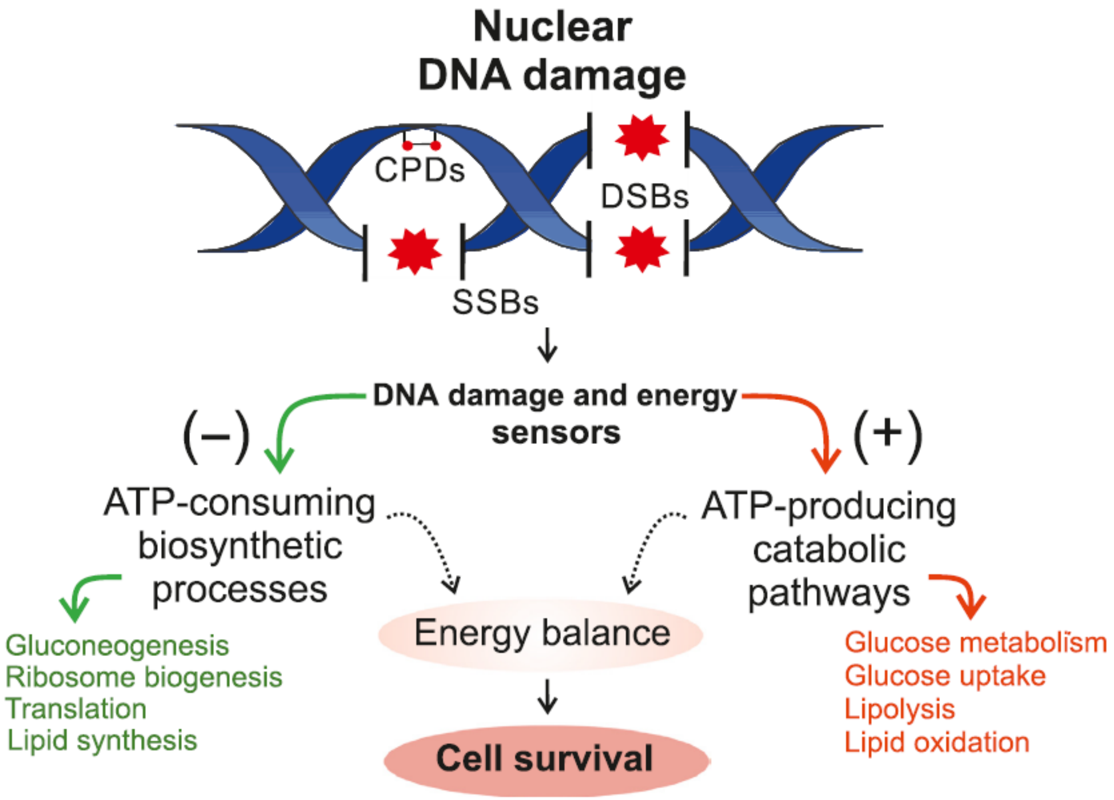

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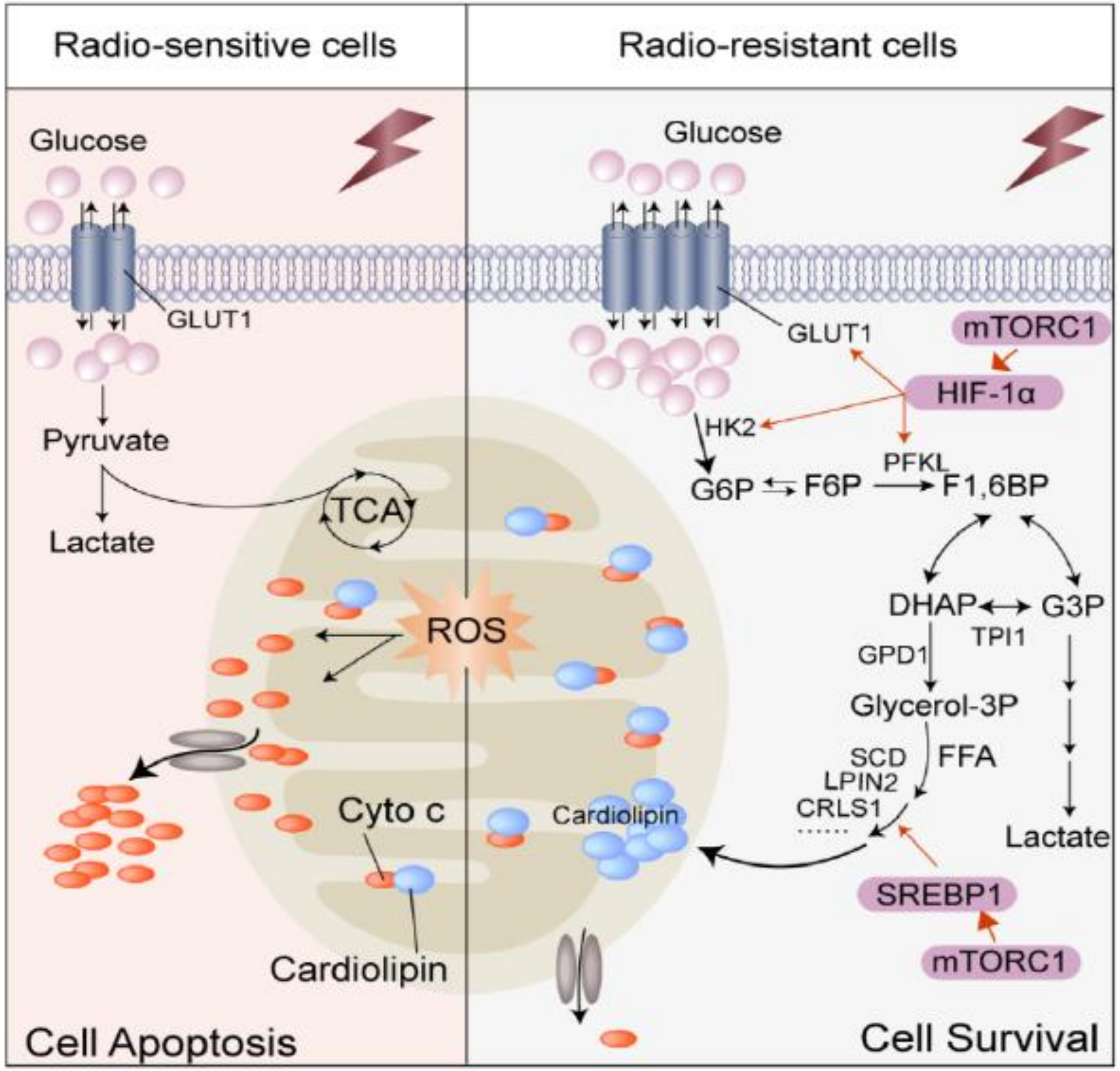

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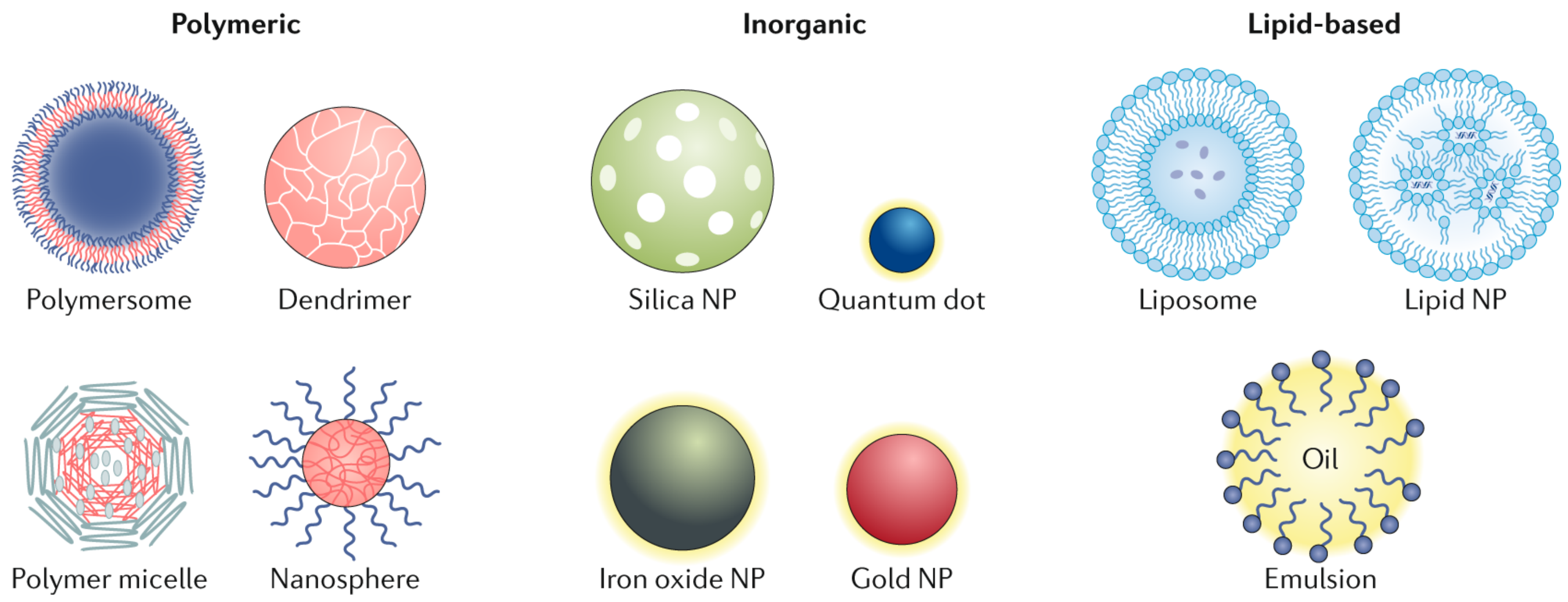
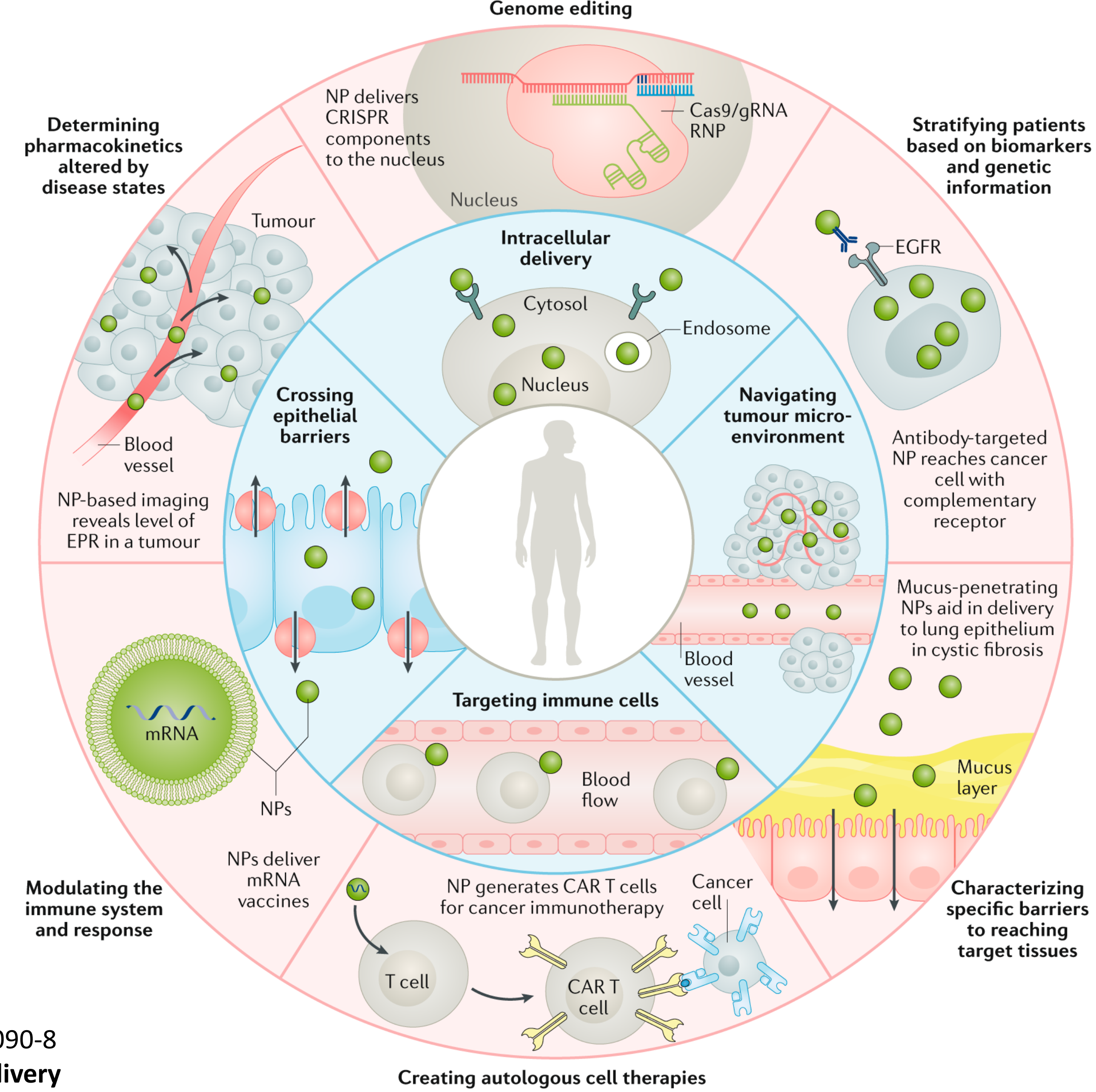


- Precise control of particle characteristics
- Payload flexibility for hydrophilic and hydrophobic cargo
- Easy surface modification
- Possibility for aggregation and toxicity

- Unique electrical, magnetic and optical properties
- Variability in size, structure and geometry
- Well suited for theranostic applications
- Toxicity and solubility limitations

- Formulation simplicity with a range of physicochemical properties
- High bioavailability
- Payload flexibility
- Low encapsulation efficiency

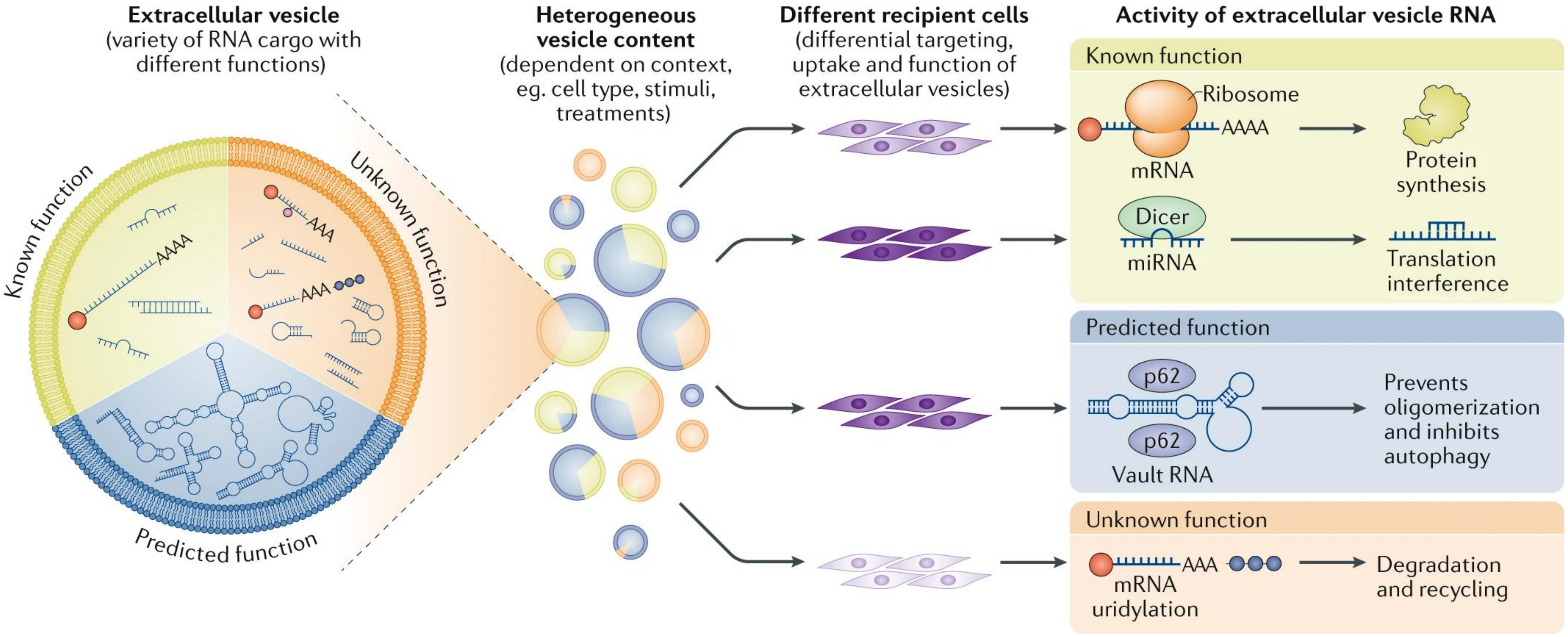
https://www.nature.com/articles/s41573-020-0090-8
Engineering precision nanoparticles for drug delivery
Nature Reviews Drug Discovery **volume 20**, pages101–124 (2021)



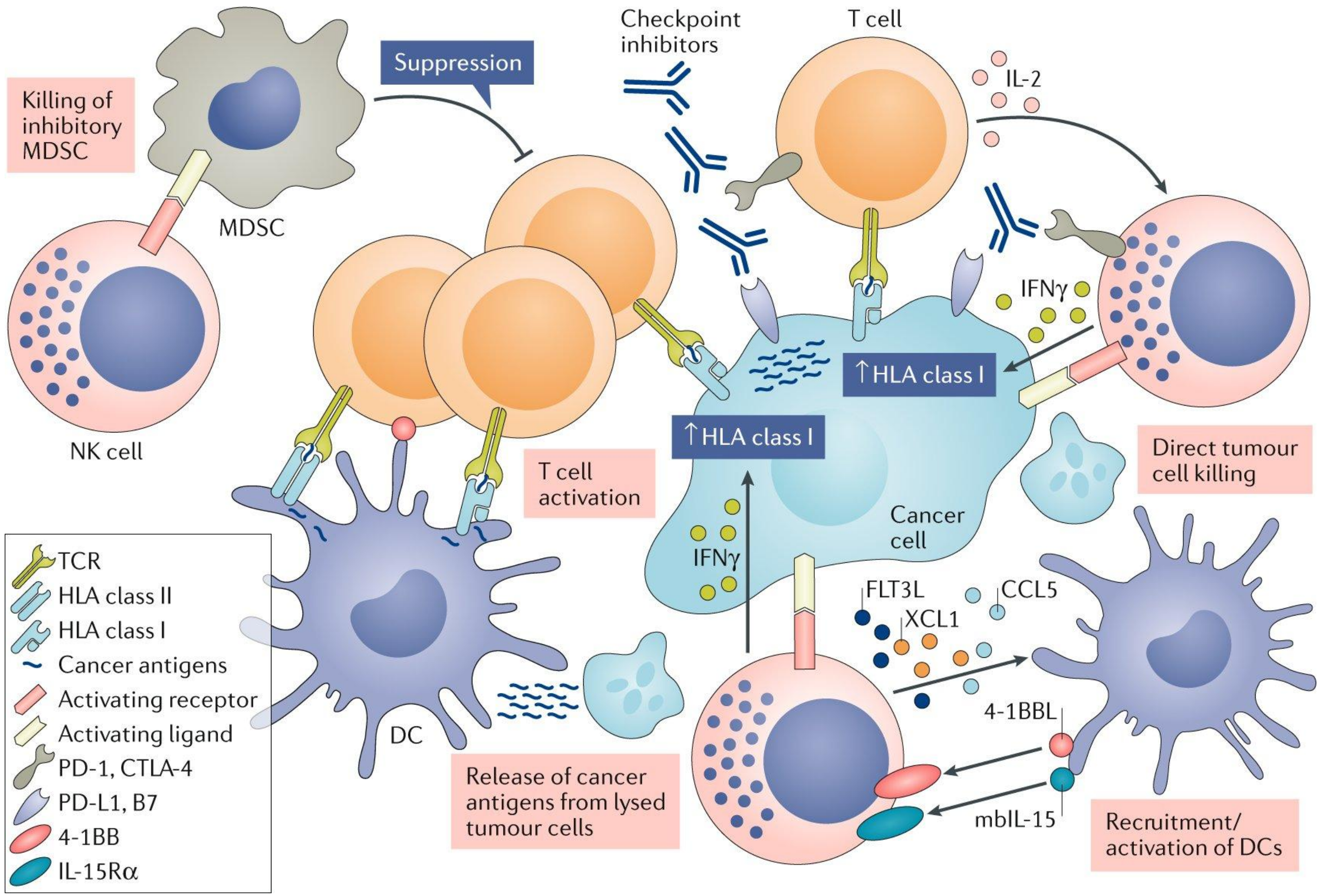
https://www.nature.com/articles/s41573-020-0090-8

Engineering precision nanoparticles for drug delivery

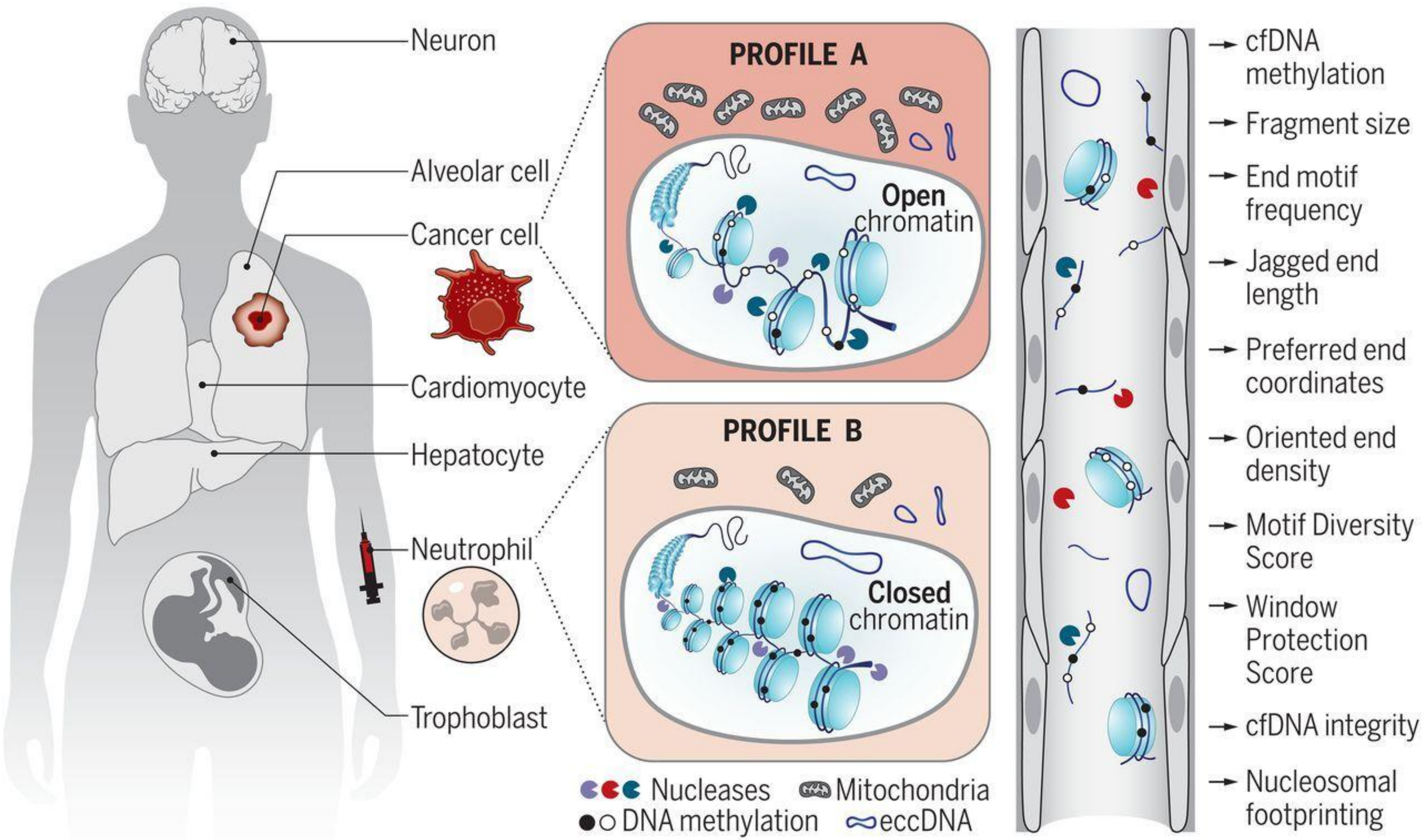
Nature Reviews Drug Discovery **volume 20**, pages101–124 (2021)



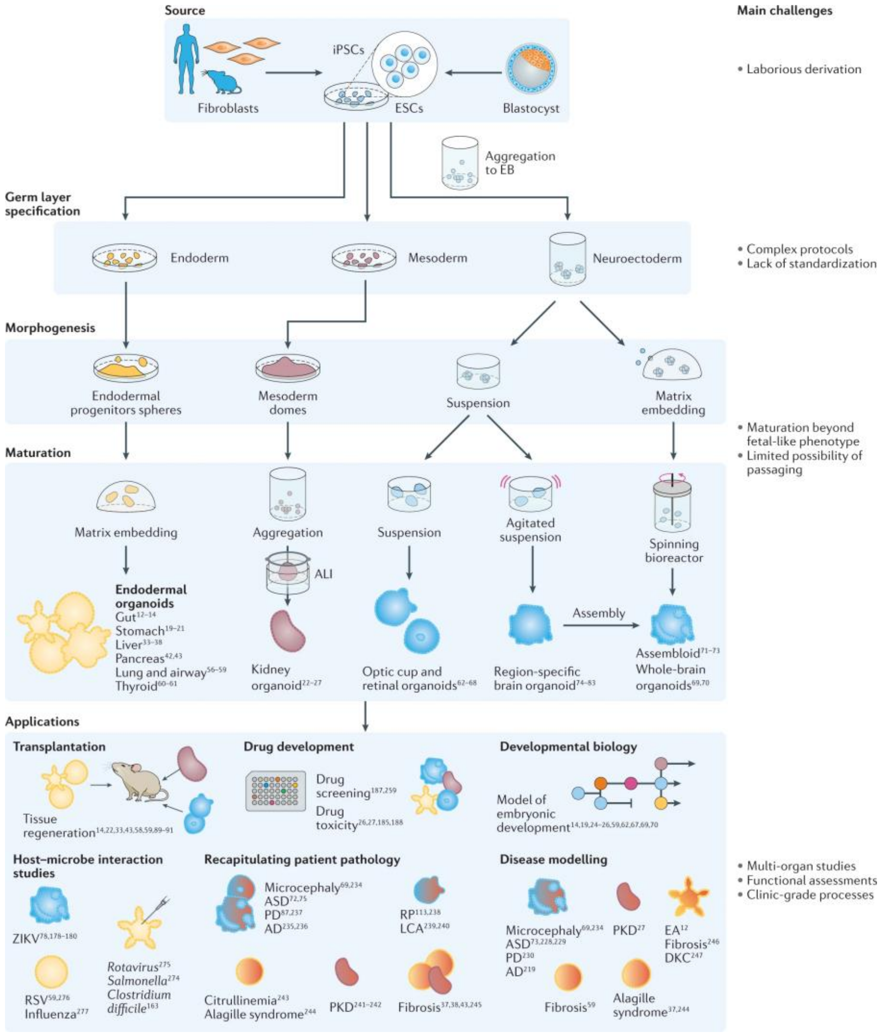
RNA delivery by extracellular vesicles in mammalian cells and its applications
Nature Reviews Molecular Cell Biology **volume 21**, pages585–606 (2020)
https://www-nature-com.libproxy1.nus.edu.sg/articles/s41580-020-0251-y

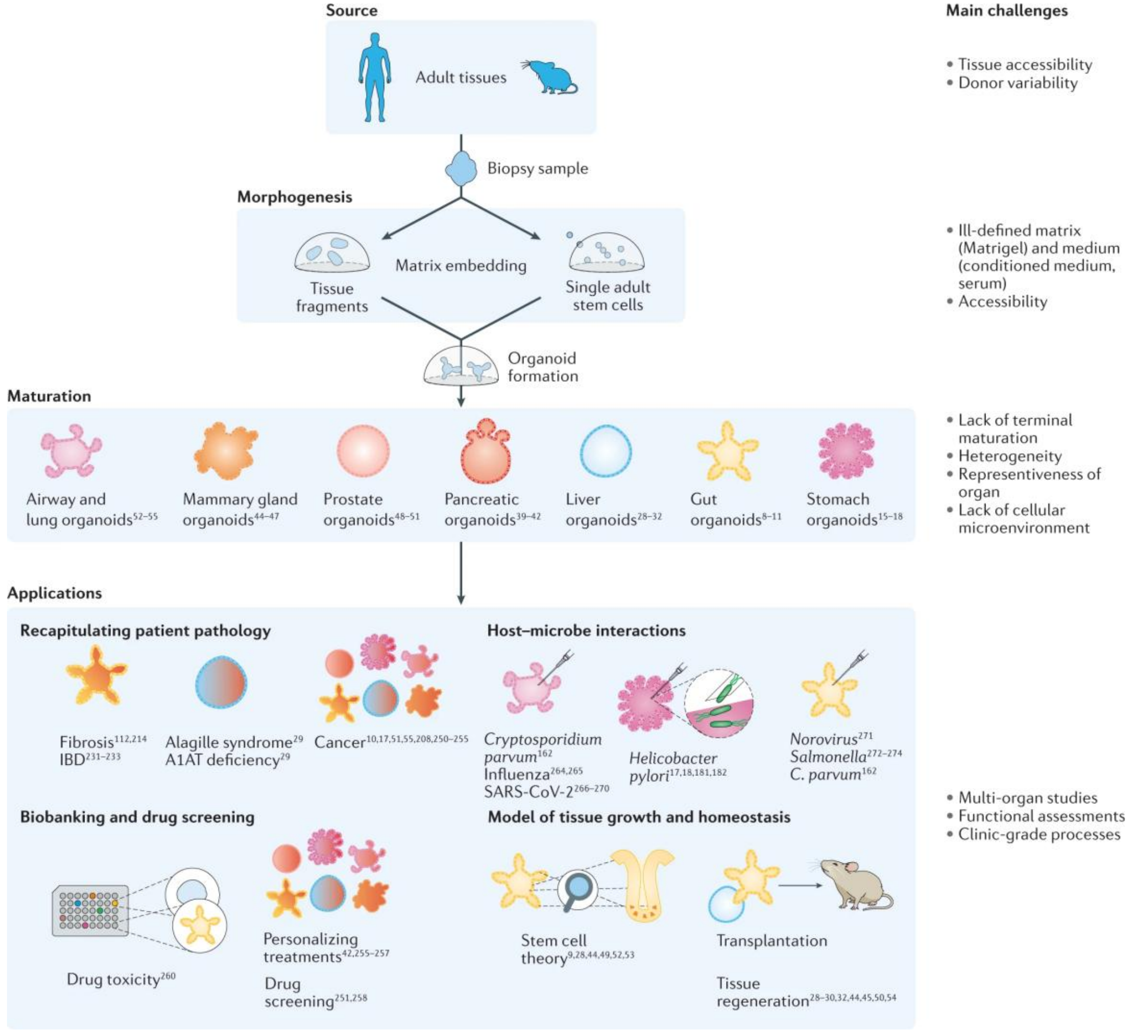


Nature Reviews Drug Discovery
https://go.nature.com/3bCLOI6
NK cells for cancer immunotherapy



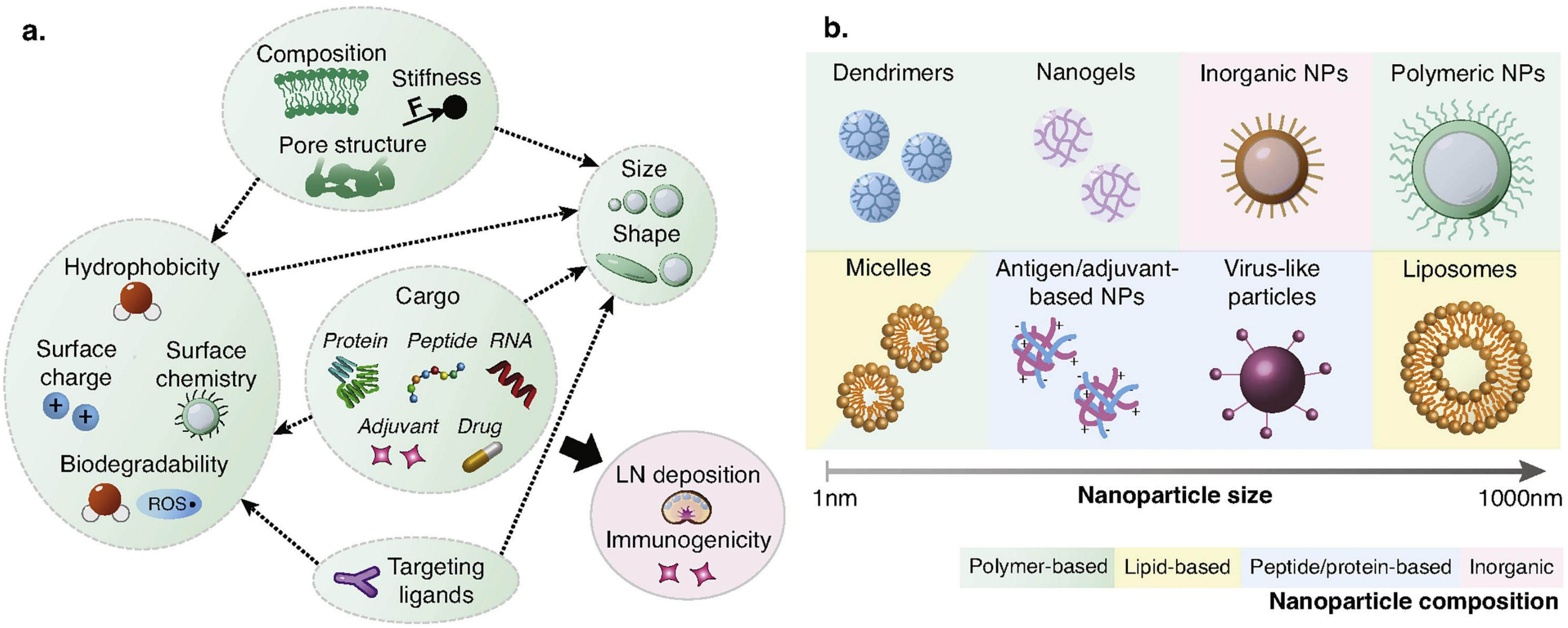
Science Magazine
https://fcld.ly/3v3uhci
Epigenetics, fragmentomics, and topology of cell-free DNA in liquid biopsies





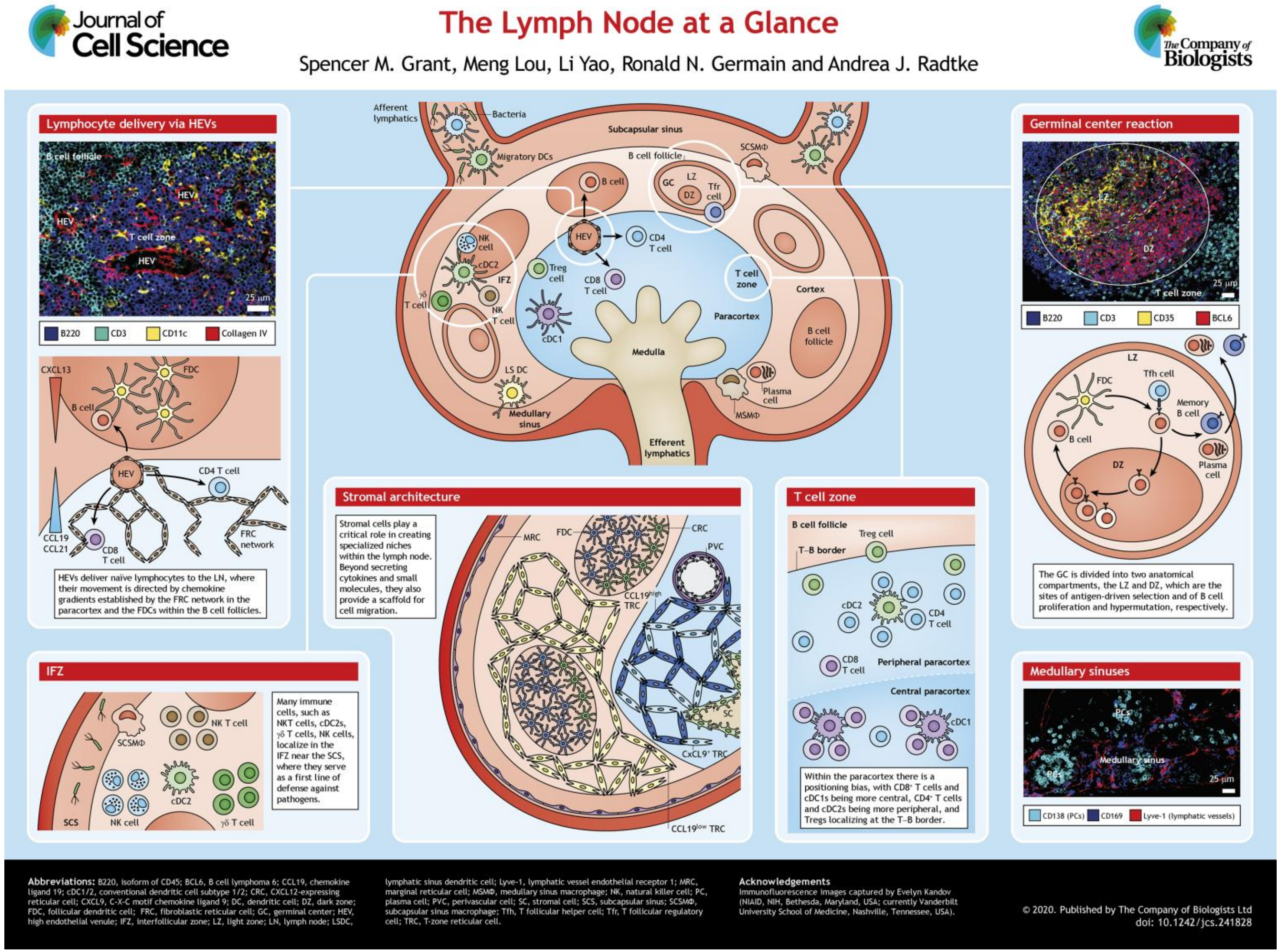
Nature Review Material
https://www.nature.com/articles/s41578-021-00279-y#Fig1
Engineering organoids

Different Stages for one technique



Advanced Drug Delivery Reviews
https://www.sciencedirect.com/science/article/pii/S0169409X20301071#f0015
Cell and tissue engineering in lymph nodes for cancer immunotherapy

Nanocarriers for lymph node targeting

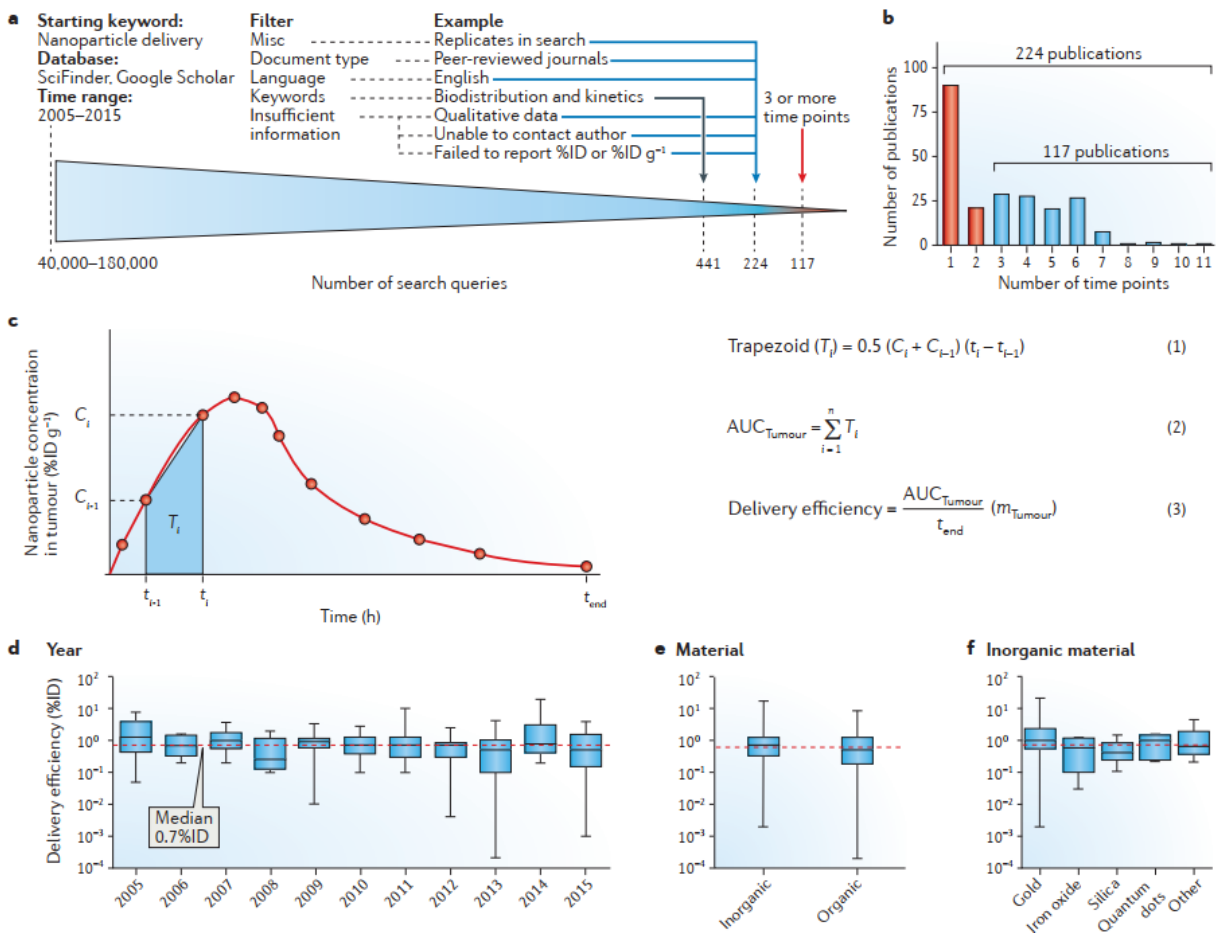


Detailed Analogy and structure of lymph node

Journal of cell Science

https://journals.biologists.com/jcs/article/133/5/jcs241828/225035/The-lymph-node-at-a-glance-how-spatial

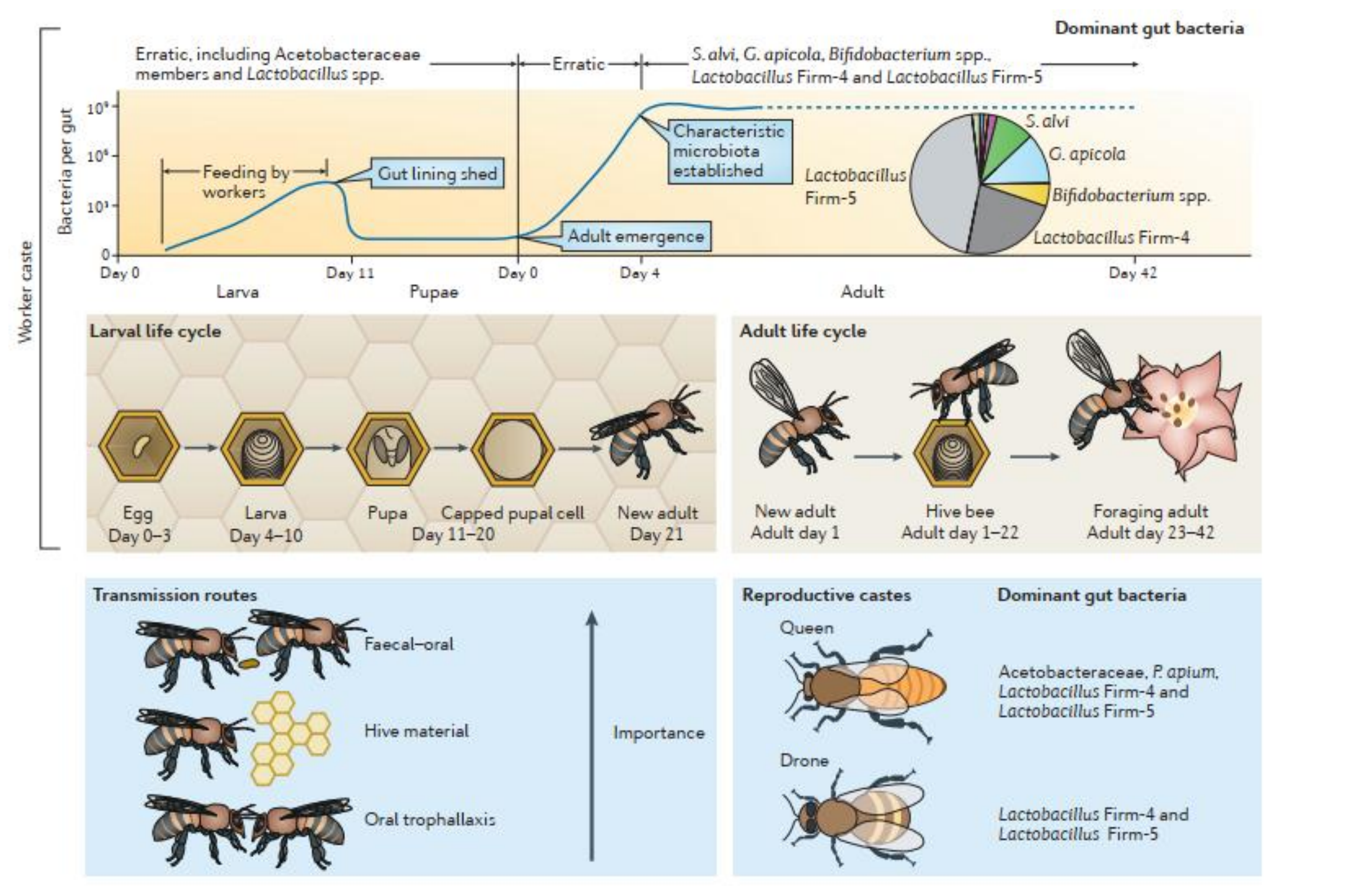
The lymph node at a glance – how spatial organization optimizes the immune response



Any comments if necessary

https://www.nature.com/articles/natrevmats201614

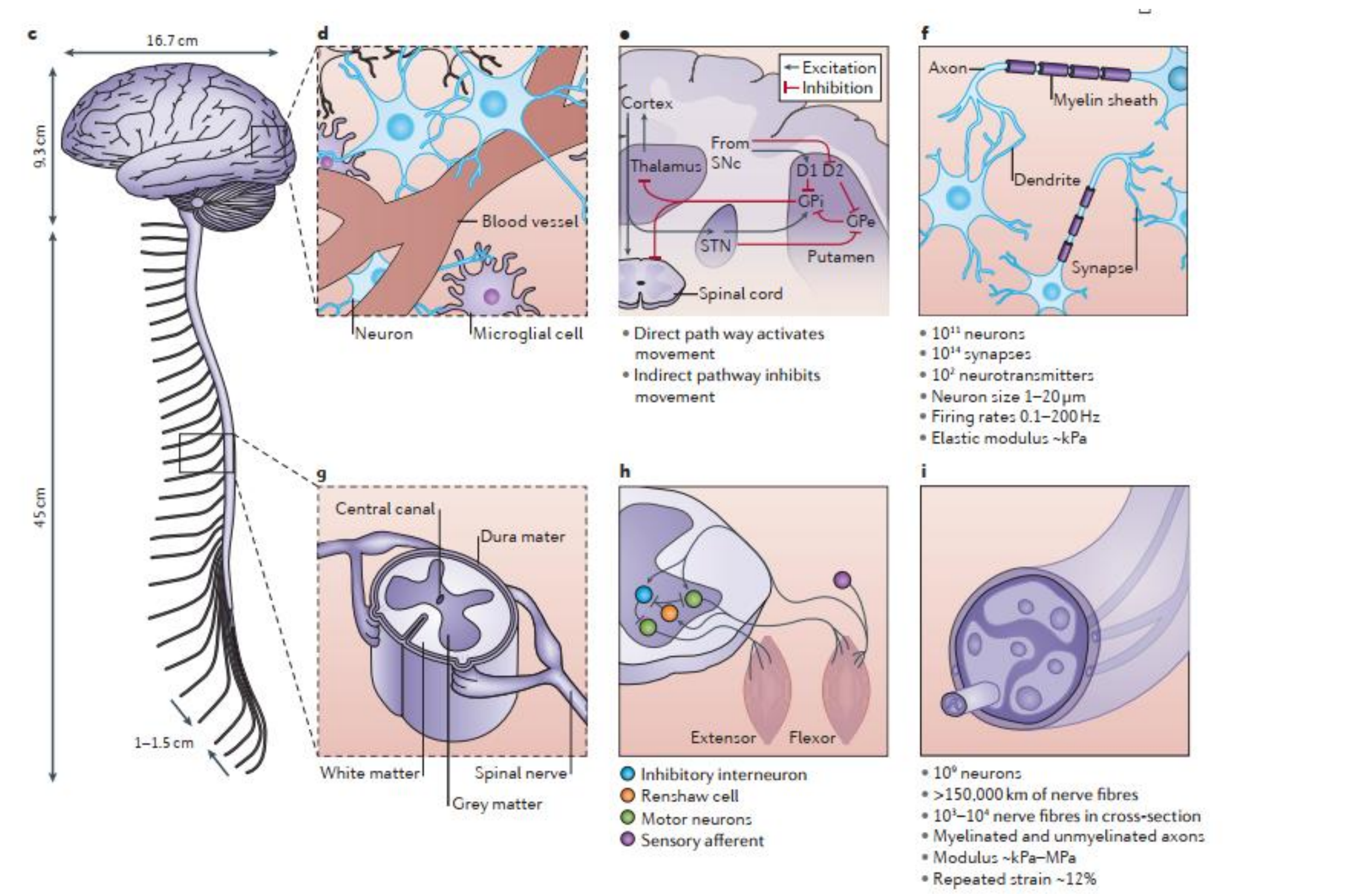
15th Aug 2021



Any comments if necessary

https://www.nature.com/articles/nrmicro.2016.43

15th Aug 2021



Any comments if necessary

https://www.nature.com/articles/natrevmats201693

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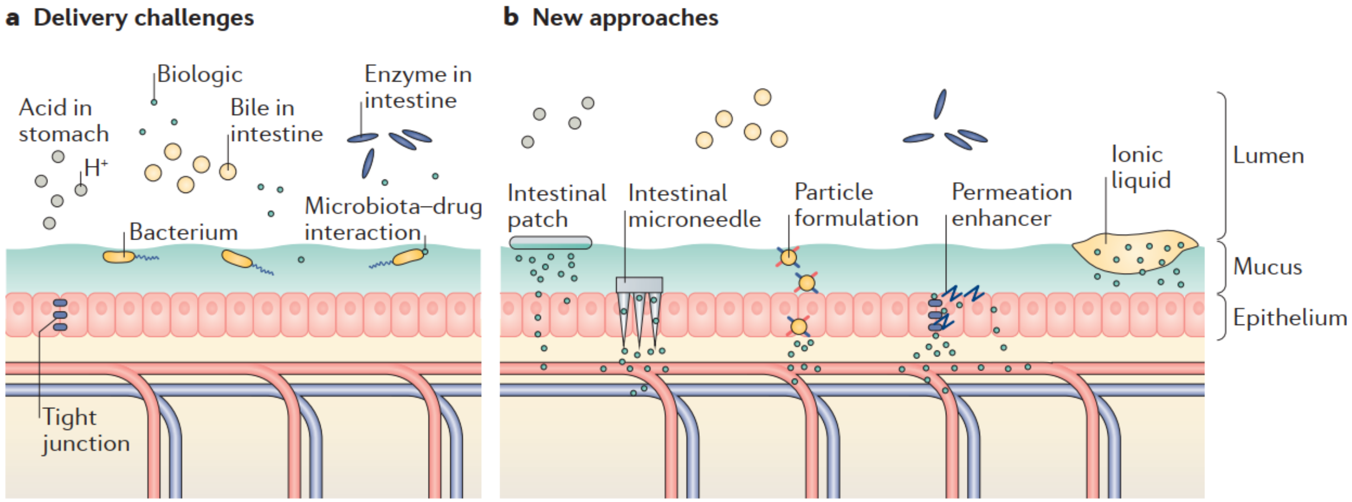
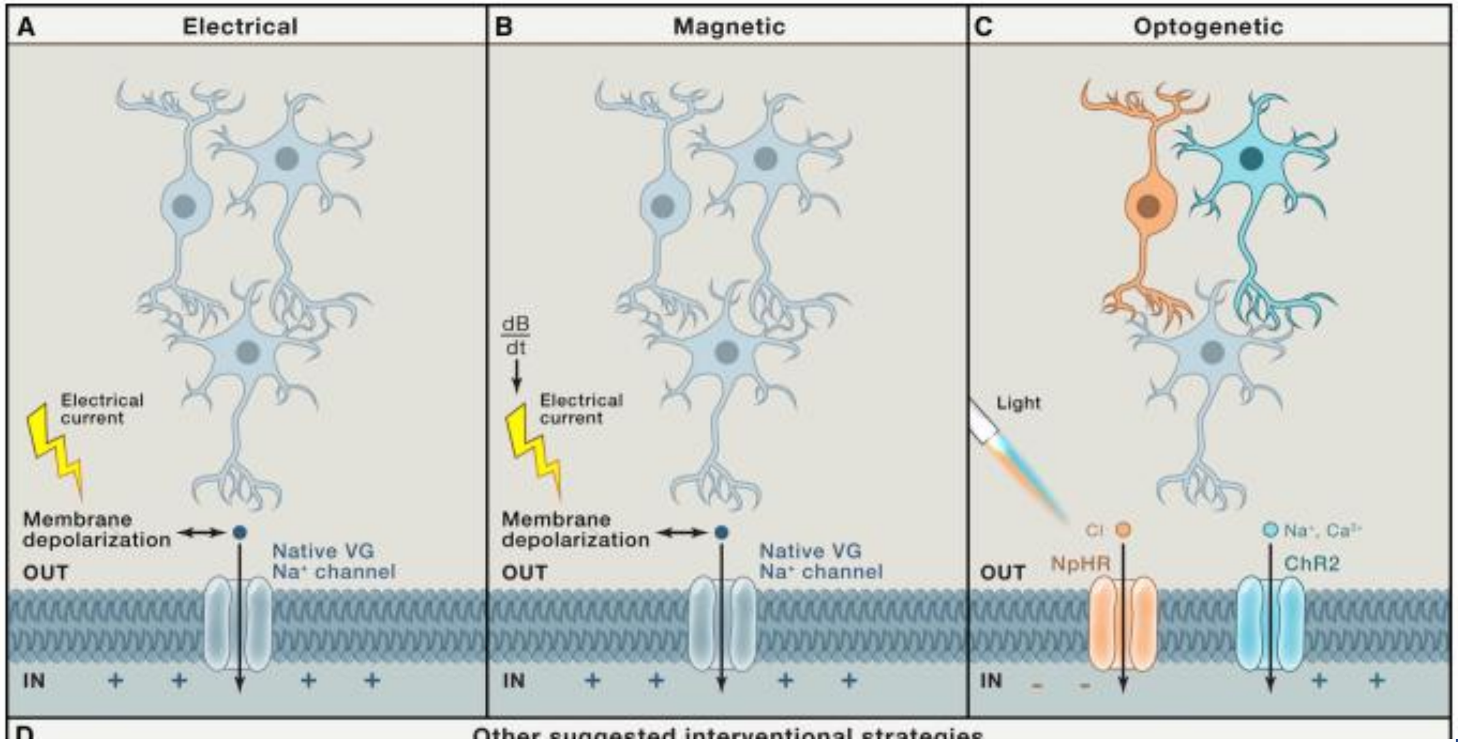


Figure 3 | **Oral barriers, products and new approaches. a** | Delivery barriers and microenvironmental challenges that limit biologic absorption via the oral route. **b** | Approaches to overcome these barriers and microenvironmental challenges. Intestinal patches improve absorption by enhancing mucoadhesion and subsequently diffusion. Intestinal microneedles can be used to facilitate absorption by physically breaching the epithelium. Particle formulations can improve transport by including intestine-targeting ligands, mucoadhesive coatings or mucopenetrative coatings.

Any comments if necessary

https://www.nature.com/articles/nrd.2018.183

15th Aug 2021



Any comments if necessary

https://pubmed.ncbi.nlm.nih.gov/27104976/

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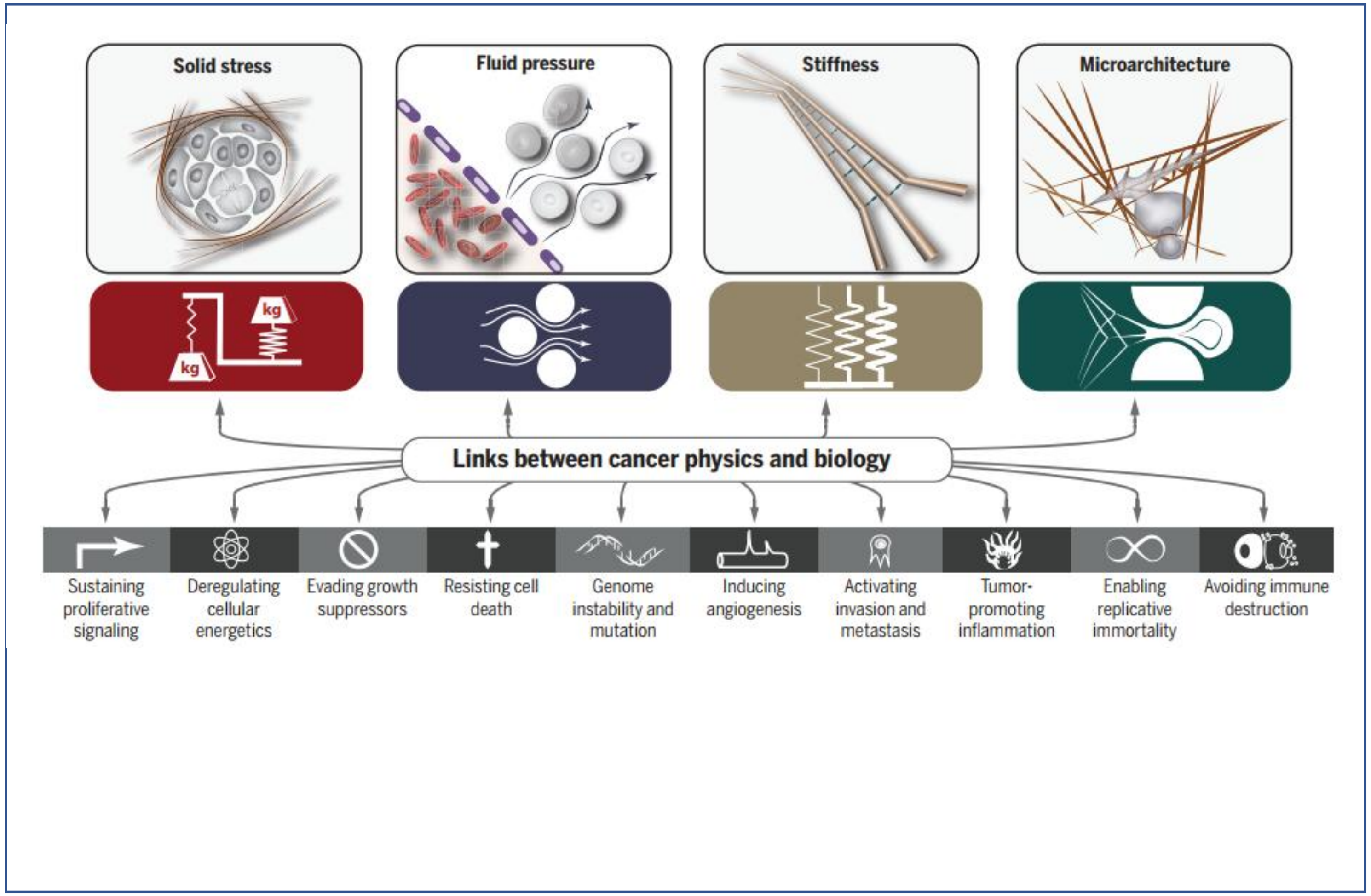
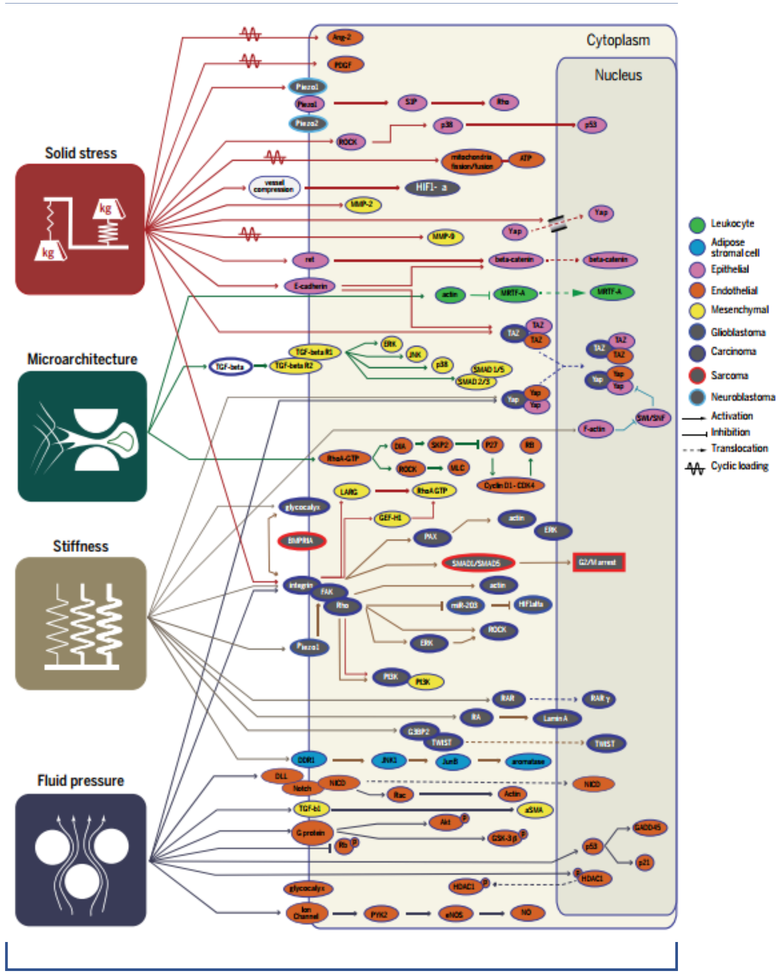
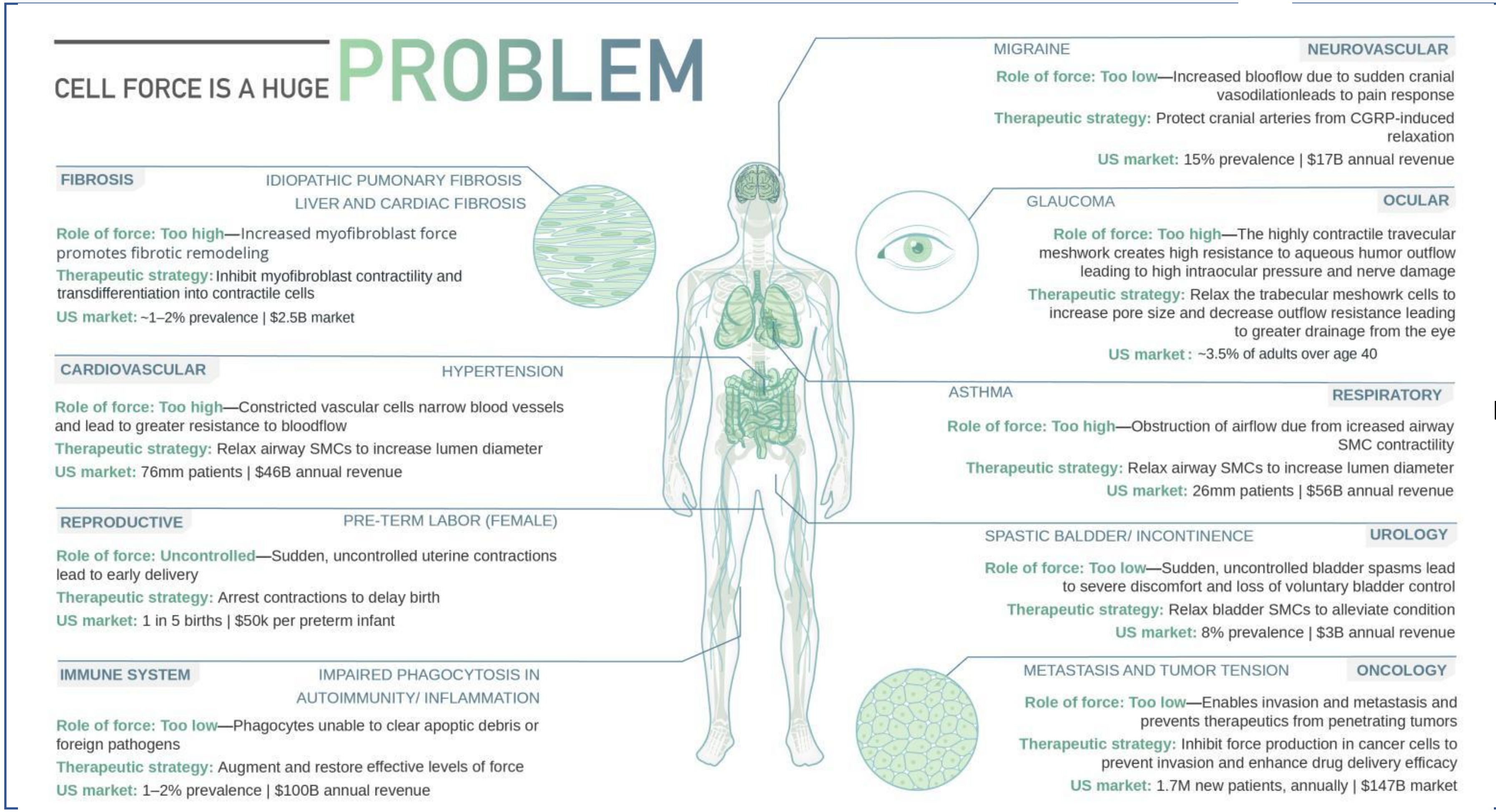




https://science.sciencemag.org/content/370/6516/eaaz0868

21st Aug 2021



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21st Aug 2021

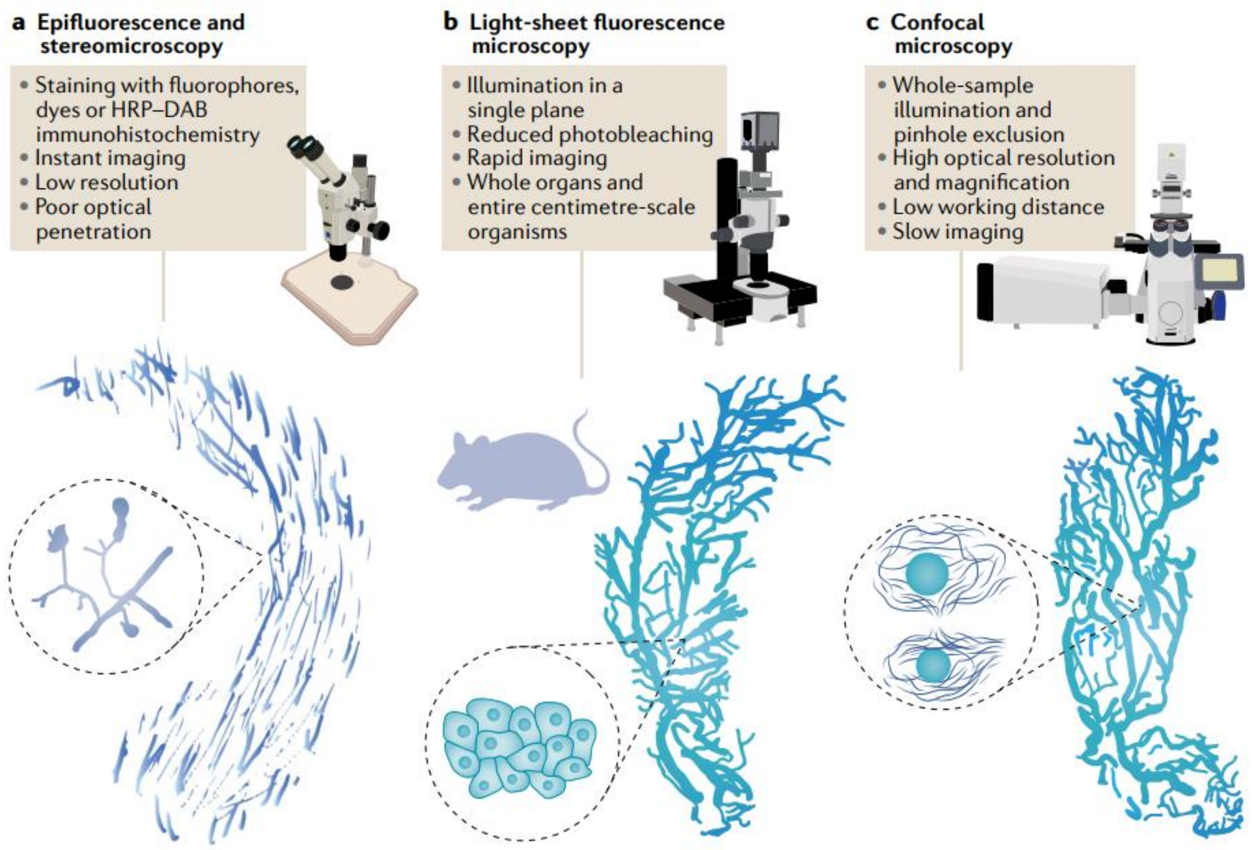
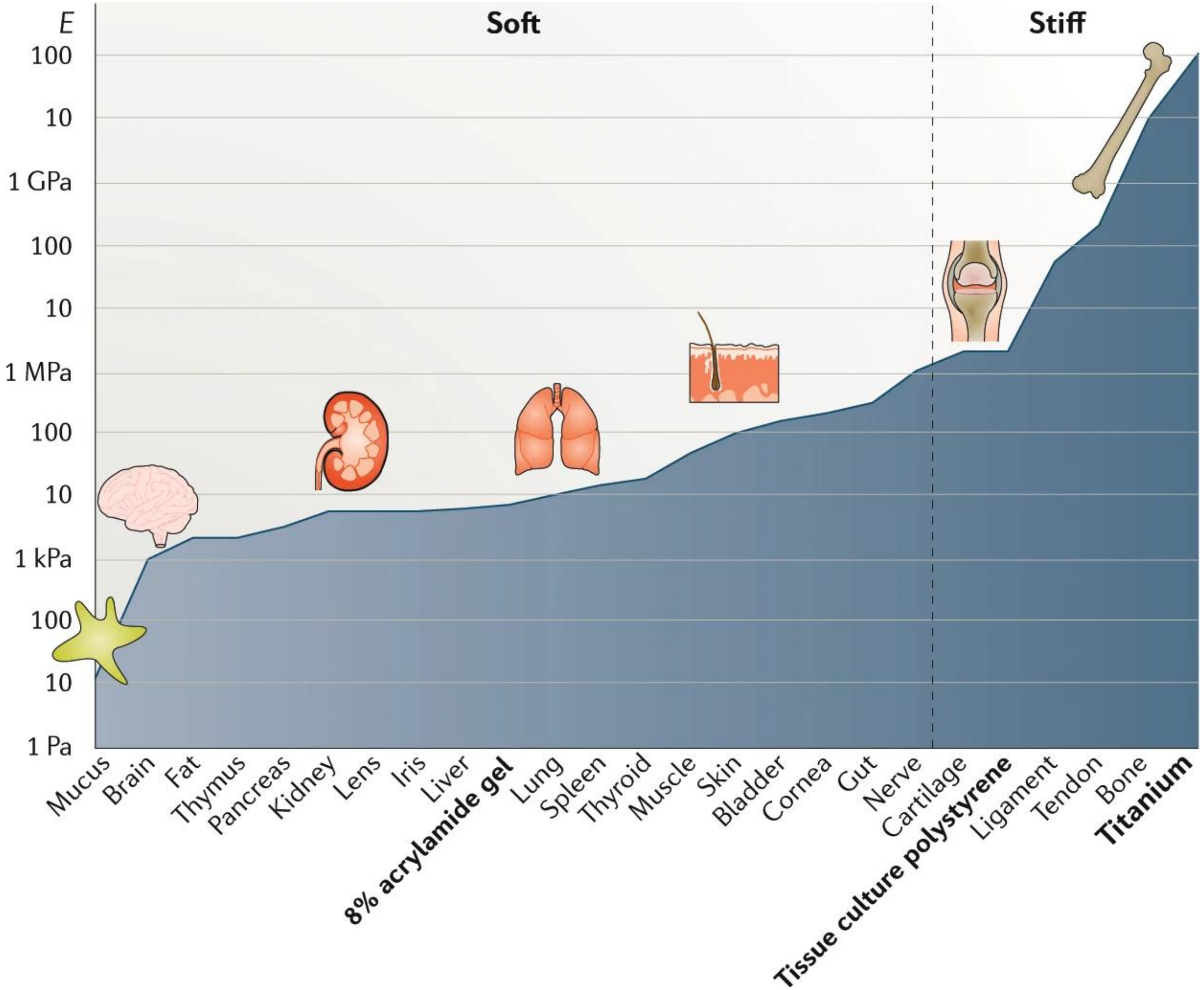


Fig. 2 | **Imaging approaches for transparent tissues. a** | Properties of epifluorescence and stereomicroscopy for cleared tissue (top); schematic of a whole cleared murine mammary gland and details of its ducts under the epifluorescence

Any comments if necessary

https://www.nature.com/articles/s41568-021-00382-w

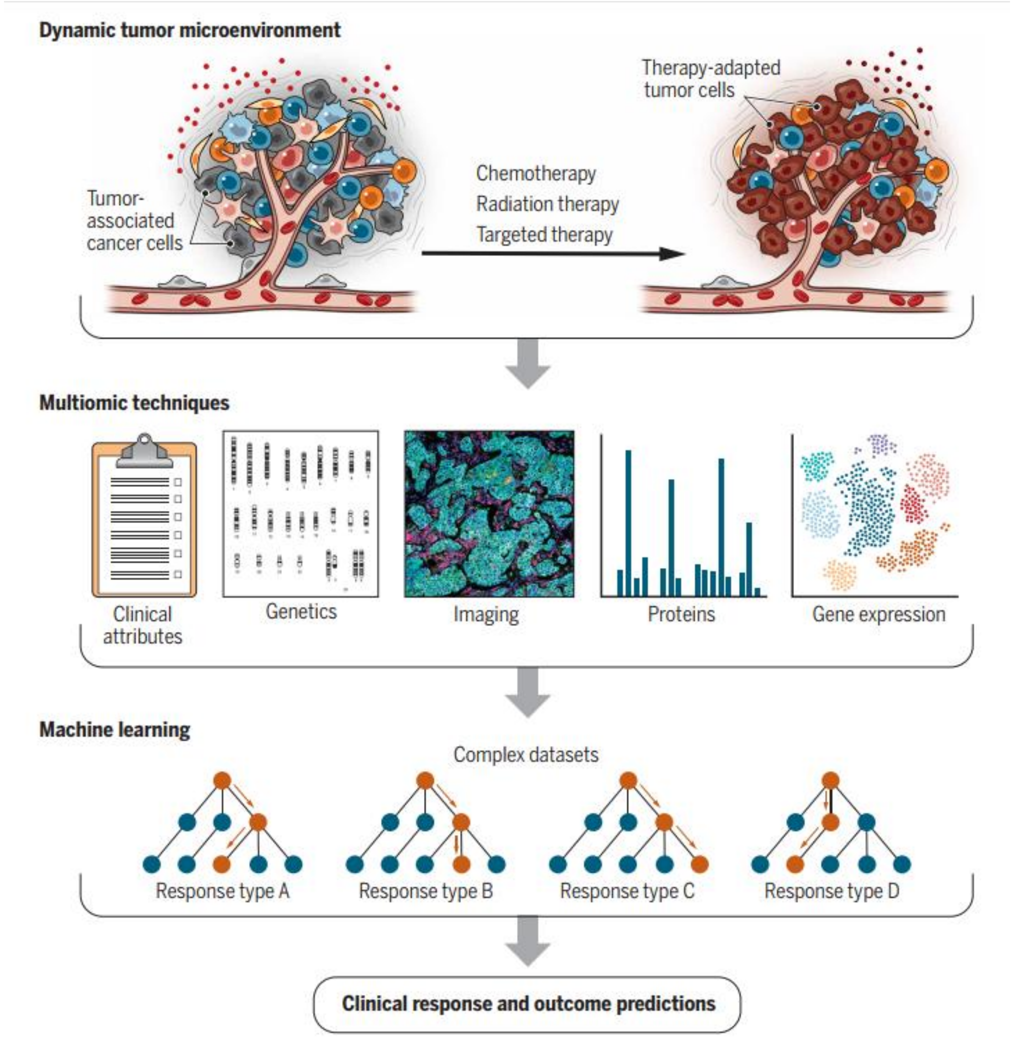
21st Aug 2021



https://www.nature.com/articles/s41578-019-0169-1?proof=t

The stiffness of living tissues and its implications for tissue engineering

6 Oct 2021

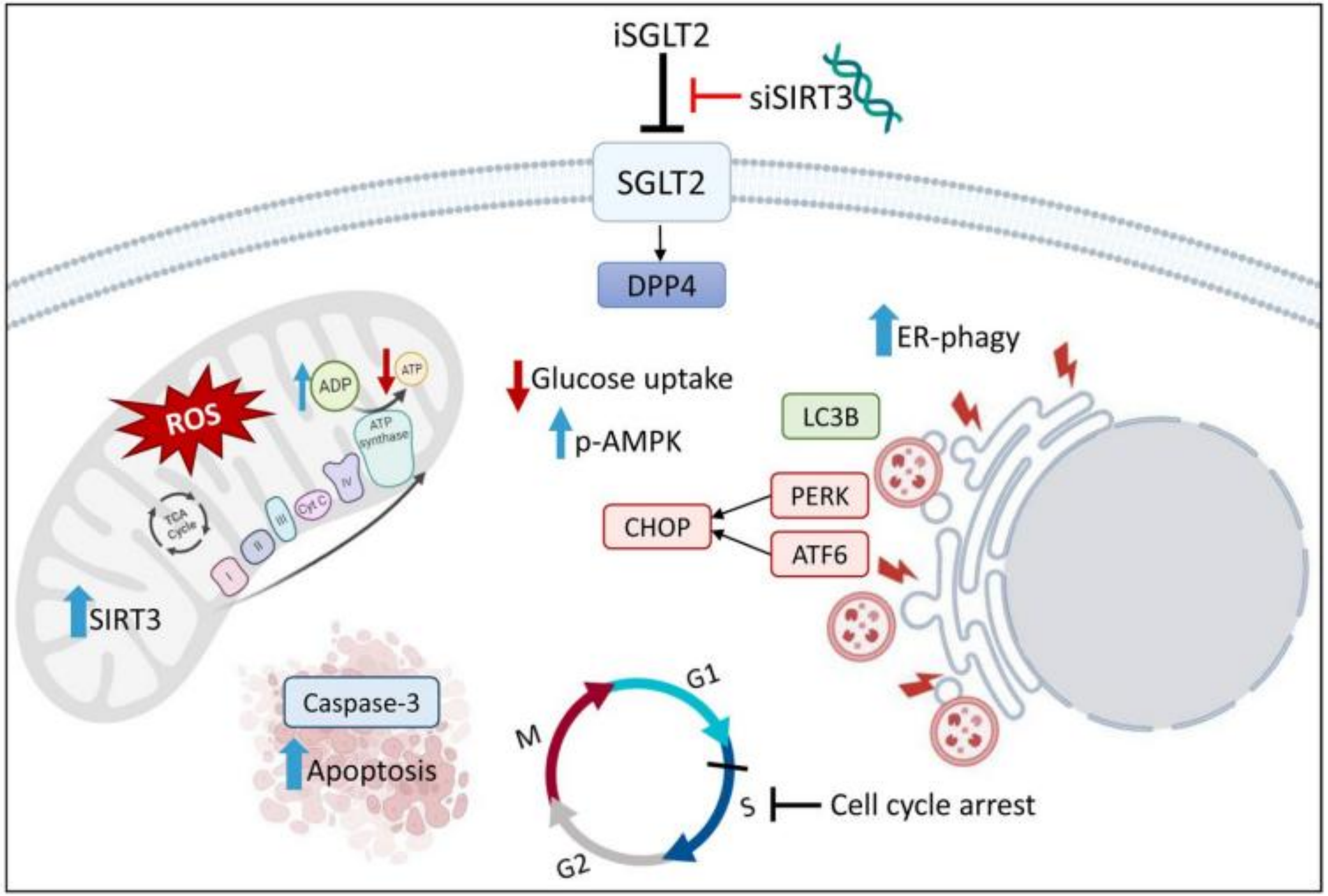


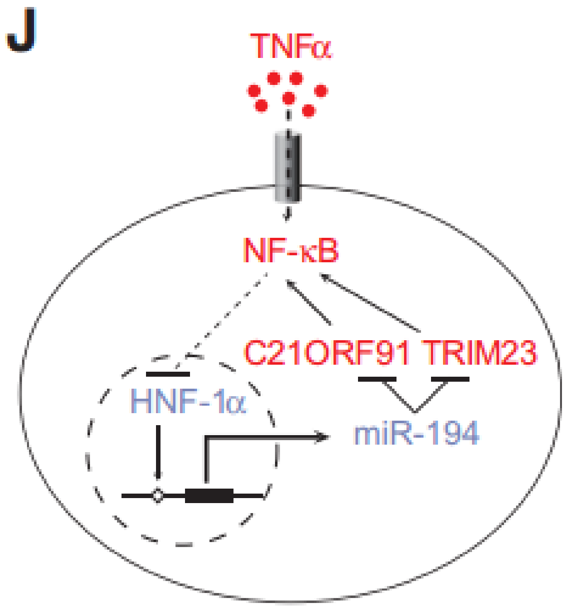
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8177696/

Decoding the dynamic tumor microenvironment

6 Oct 2021

Graphical Abstract





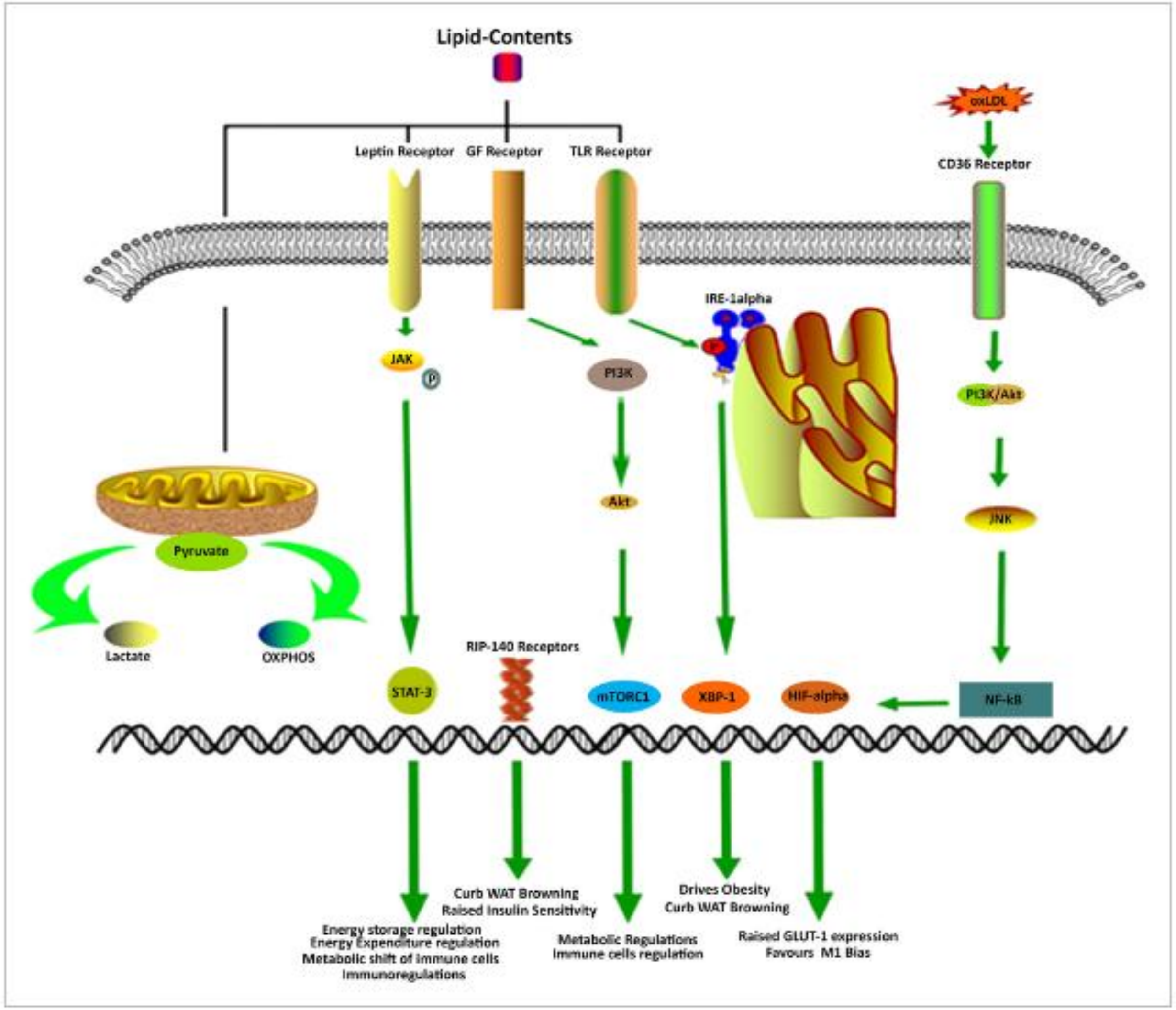
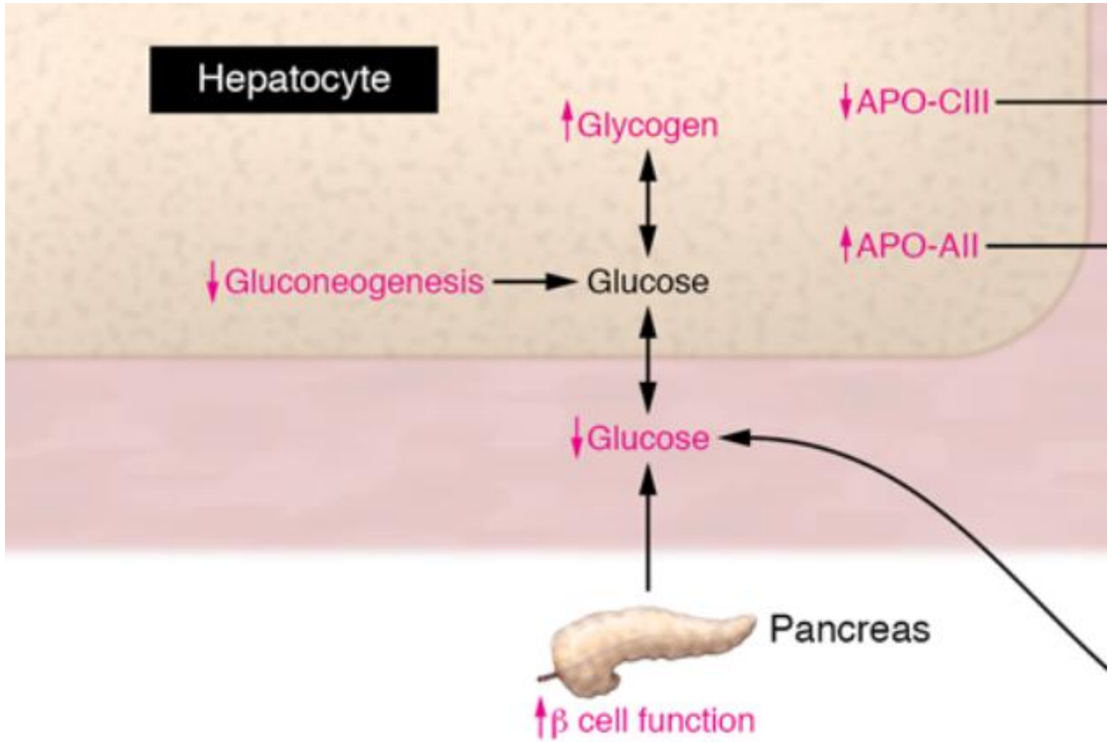
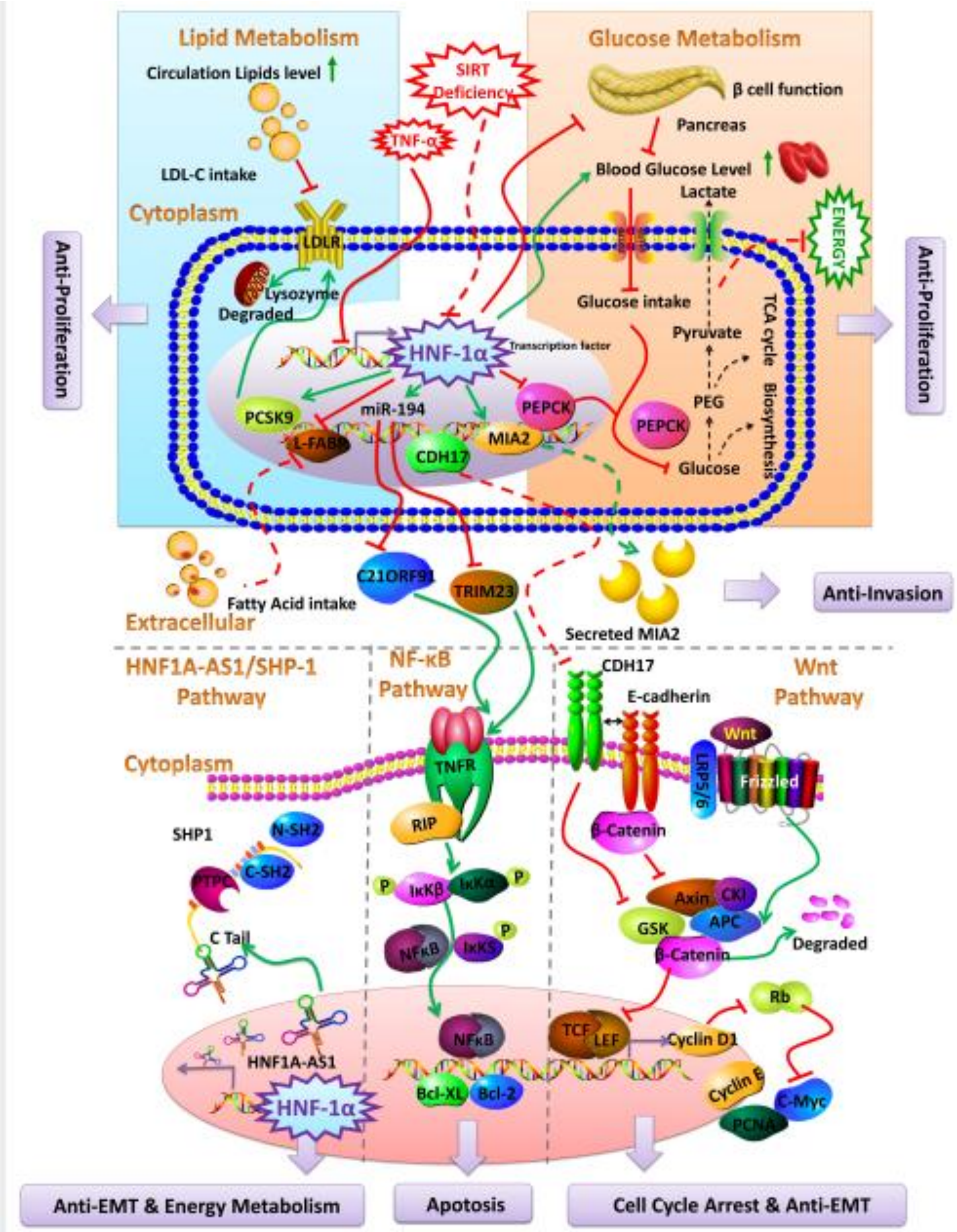


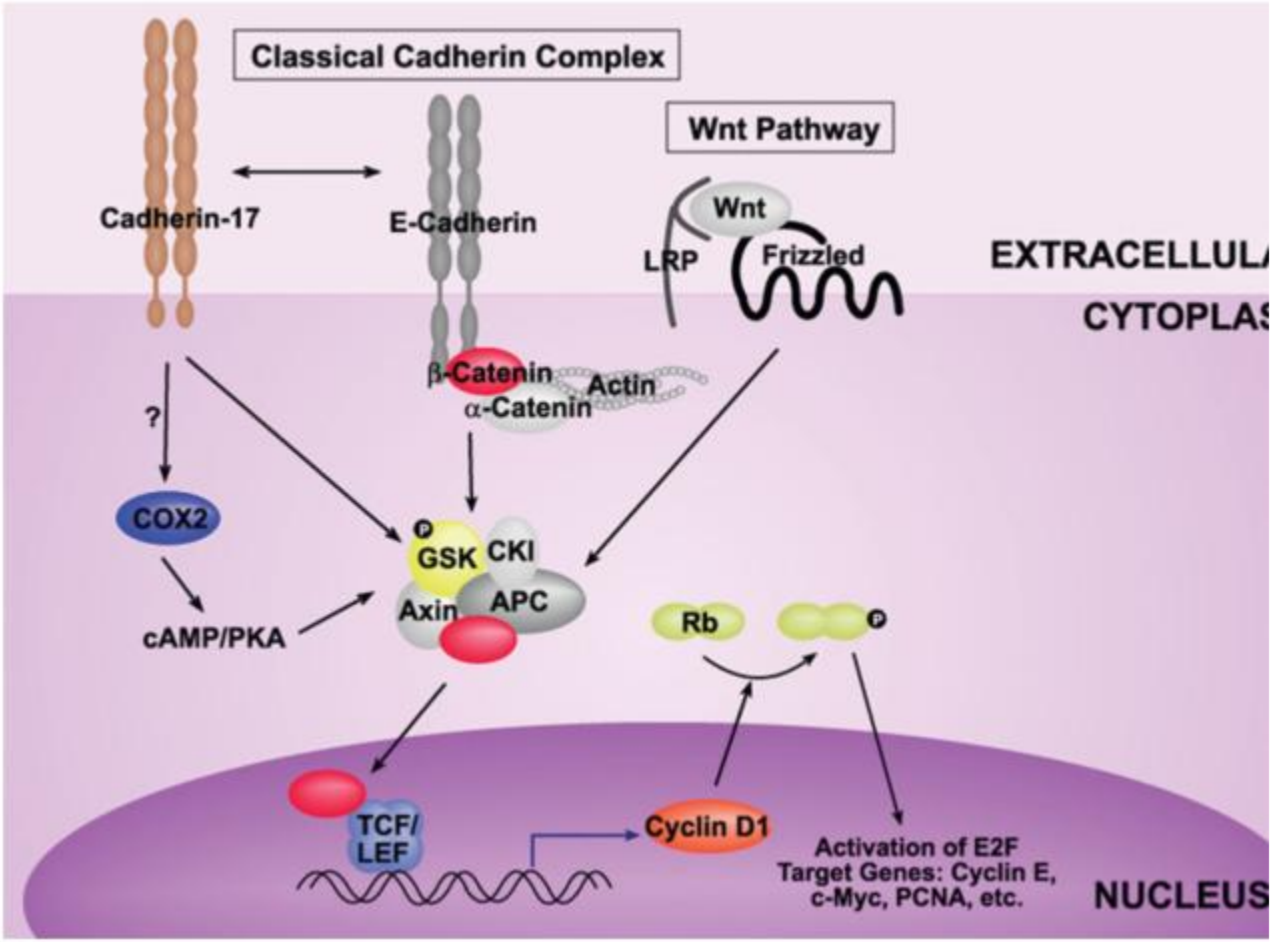
Figure 1

Open in figure viewer | PowerPoint

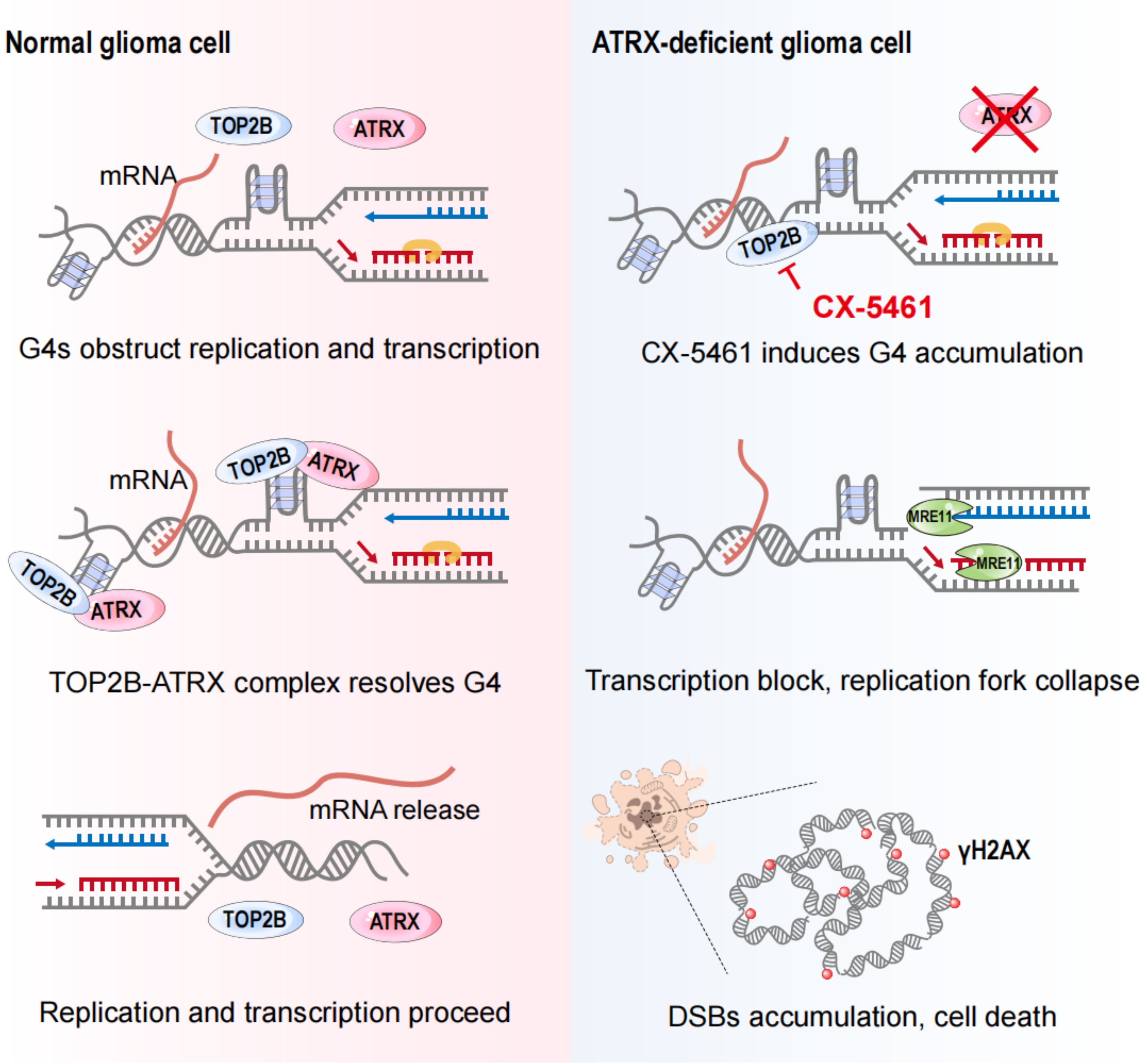
Molecular interactions in Immunometabolism governing lipid metabolic disorders. Induction of Leptin receptor, GF receptors, TLR receptors and CD36 receptors lead to the activation of STAT3, mTORC1, XBP − 1 (through IRE − 1α) and HIF − 1α that steer to interference in lipid

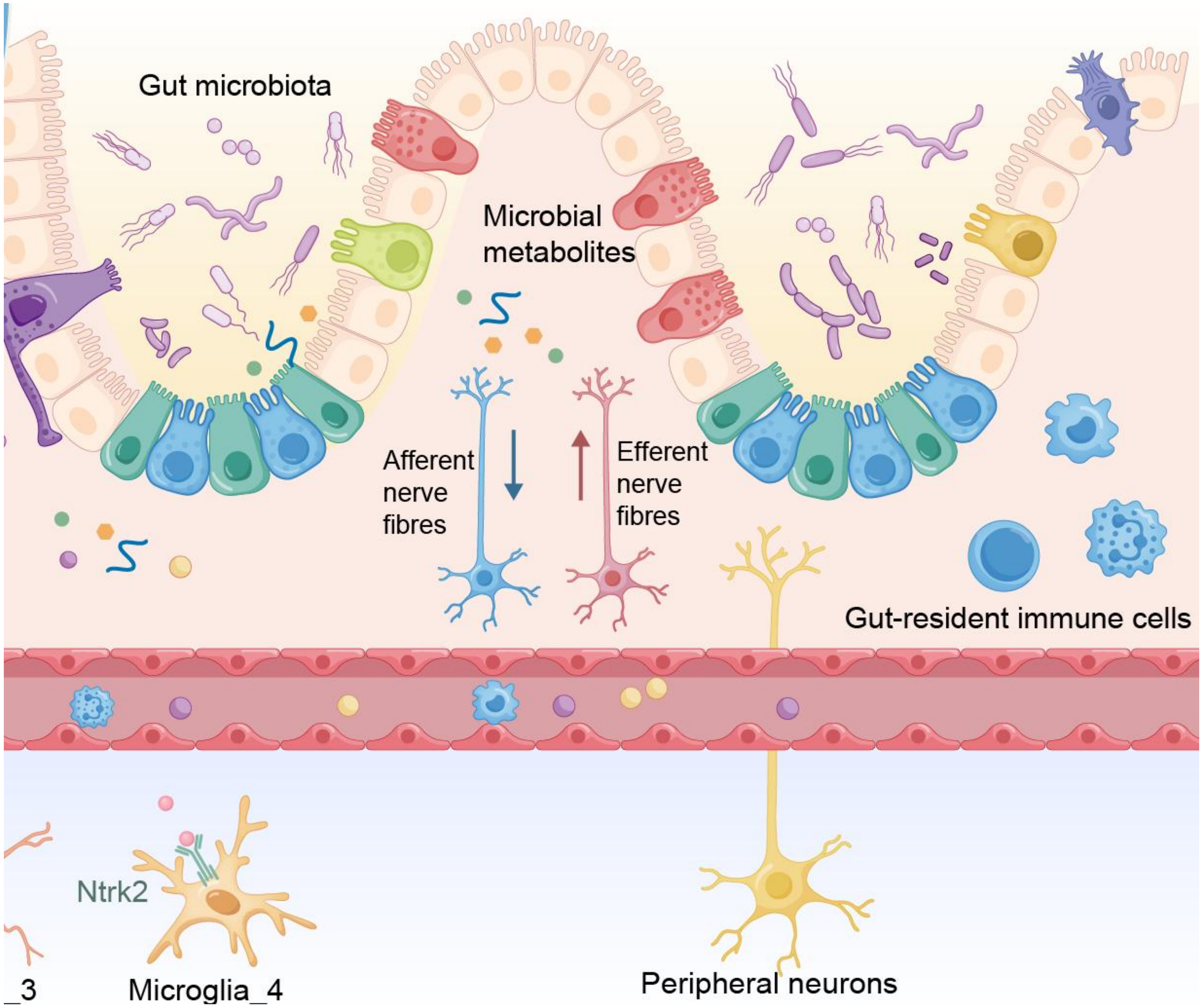


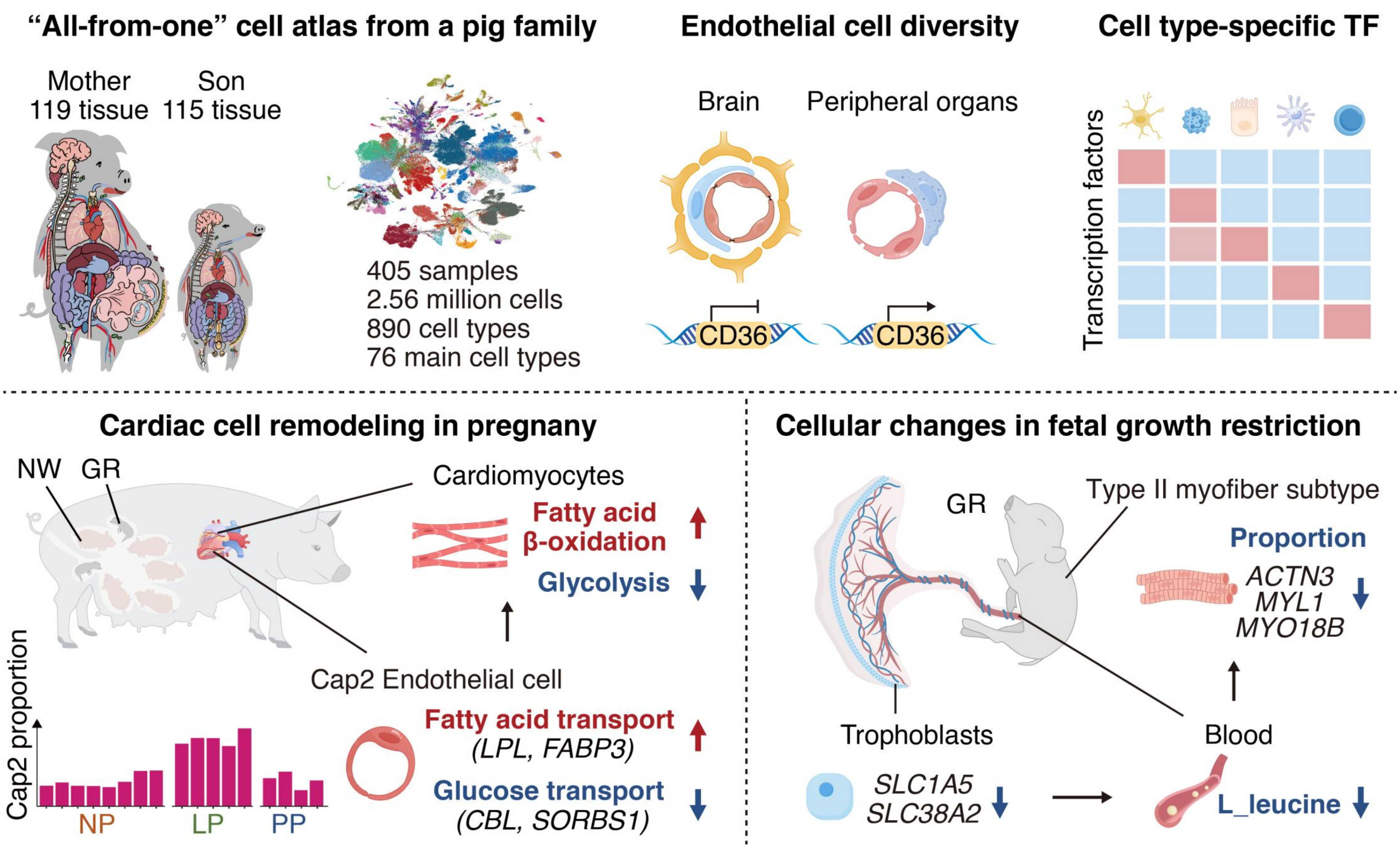

Lipid Metabolism
Circulation Lipids level
LDL-C intake
Cytoplasm
LDLR
Lysozyme
Degraded
SIRT Deficiency
TNF-α
Glucose Metabolism
β cell function
Pancreas
Blood Glucose Level
Lactate
ENERGY
Glucose intake
Pyruvate
TCA cycle
PEG
Biosynthesis
PEPCK
Glucose
HNF-1α
Transcription factor
PCSK9
miR-194
L-FABP
CDH17
MIA2
Anti-Proliferation
Anti-Proliferation
Fatty Acid intake
Extracellular
C21ORF91
TRIM23
Secreted MIA2
Anti-Invasion
HNF1A-AS1/SHP-1 Pathway
NF-κB Pathway
Wnt Pathway
CDH17
E-cadherin
Cytoplasm
TNFR
RIP
IκKβ
IκKα
NFκB
IκKS
SHP1
N-SH2
C-SH2
PTPC
C Tail
LRP5/6
Frizzled
Wnt
β-Catenin
Axin
CKI
GSK
APC
β-Catenin
Degraded
Rb
TCF
LEF
Cyclin D1
Cyclin E
C-Myc
PCNA
NFκB
Bcl-XL
Bcl-2
HNF1A-AS1
HNF-1α
Anti-EMT & Energy Metabolism
Apoptosis
Cell Cycle Arrest & Anti-EMT

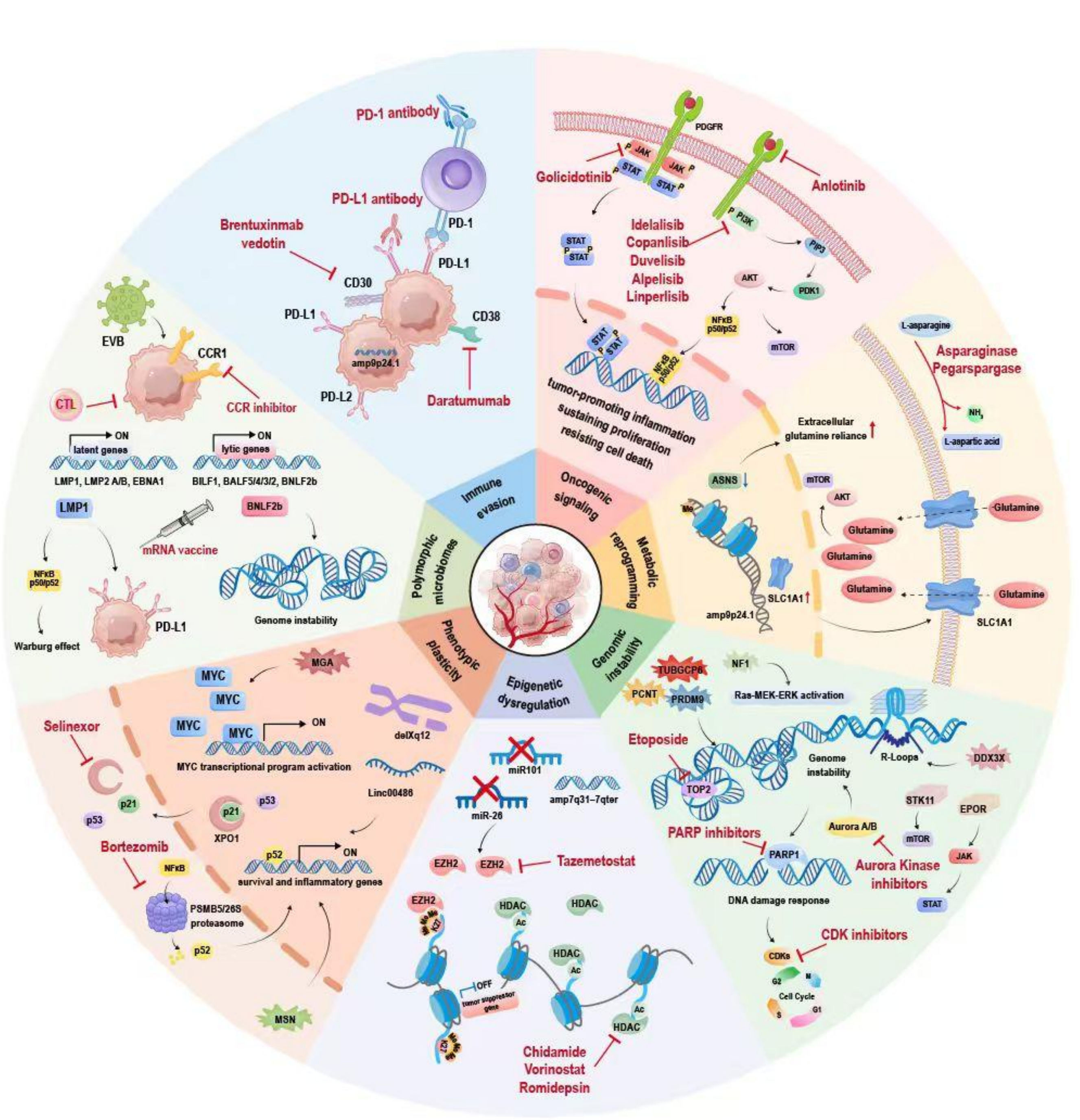

Classical Cadherin Complex
Wnt Pathway
Cadherin-17
E-Cadherin
Wnt
LRP
Frizzled
EXTRACELLULA
CYTOPLAS
β-Catenin
Actin
α-Catenin
?
COX2
cAMP/PKA
GSK
CKI
Axin
APC
Rb
TCF/
LEF
Cyclin D1
Activation of E2F
Target Genes: Cyclin E,
c-Myc, PCNA, etc.
NUCLEUS

Graphical abstract




Gut microbiota
Microbial metabolites
Afferent nerve fibres
Efferent nerve fibres
Gut-resident immune cells
Ntrk2
_3
Microglia_4
Peripheral neurons


"All-from-one" cell atlas from a pig family
Mother
119 tissue
Son
115 tissue
405 samples
2.56 million cells
890 cell types
76 main cell types
Endothelial cell diversity
Brain
Peripheral organs
CD36
CD36
Cell type-specific TF
Transcription factors
Cardiac cell remodeling in pregnany
NW
GR
Cardiomyocytes
Fatty acid β-oxidation
Glycolysis
Cap2 Endothelial cell
Fatty acid transport
(LPL, FABP3)
Glucose transport
(CBL, SORBS1)
Cap2 proportion
NP
LP
PP
Cellular changes in fetal growth restriction
GR
Type II myofiber subtype
Proportion
ACTN3
MYL1
MYO18B
Trophoblasts
Blood
SLC1A5
SLC38A2
L_leucine



Immune evasion
Oncogenic signaling
Metabolic reprogramming
Genomic instability
Epigenetic dysregulation
Phenotypic plasticity
Polymorphic microbiomes
PD-1 antibody
PD-L1 antibody
PD-1
PD-L1
Brentuxinmab vedotin
CD30
CD38
PD-L1
amp9p24.1
PD-L2
Daratumumab
PDGFR
JAK
STAT
Golicidotinib
Anlotinib
Idelalisib
Copanlisib
Duvelisib
Alpelisib
Linperlisib
PI3K
PIP3
PDK1
AKT
NFκB p50/p52
mTOR
tumor-promoting inflammation
sustaining proliferation
resisting cell death
L-asparagine
Asparaginase
Pegarspargase
NH3
L-aspartic acid
Extracellular glutamine reliance
ASNS
mTOR
AKT
Glutamine
Glutamine
Glutamine
Glutamine
Glutamine
Glutamine
SLC1A1
amp9p24.1
SLC1A1
TUBGCP6
NF1
PCNT
PRDM9
Ras-MEK-ERK activation
Etoposide
TOP2
Genome instability
R-Loops
DDX3X
STK11
EPOR
mTOR
JAK
STAT
Aurora A/B
Aurora Kinase inhibitors
PARP inhibitors
PARP1
DNA damage response
CDK inhibitors
CDKs
G2
M
Cell Cycle
S
G1
miR101
miR-26
amp7q31–7qter
EZH2
EZH2
Tazemetostat
EZH2
HDAC
Ac
HDAC
HDAC
Ac
OFF
tumor suppressor gene
HDAC
Ac
Chidamide
Vorinostat
Romidepsin
MGA
MYC
MYC
MYC
MYC
ON
delXq12
MYC transcriptional program activation
Linc00486
Selinexor
p21
p53
p21
XPO1
p53
Bortezomib
NFκB
p52
ON
survival and inflammatory genes
PSMB5/26S proteasome
p52
MSN
EVB
CCR1
CTL
CCR inhibitor
ON
latent genes
ON
lytic genes
LMP1, LMP2 A/B, EBNA1
BILF1, BALF5/4/3/2, BNLF2b
LMP1
BNLF2b
mRNA vaccine
NFκB p50/p52
PD-L1
Genome instability
Warburg effect